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ROBOTIC SURGICAL TOOL WITH REPLACEABLE CARRIAGE

Abstract

A surgical tool includes a stage portion that includes opposing first and second ends, a lead screw and at least one spline that extend between the first and second ends, and a first layer of a carriage movably mounted to the lead screw and the at least one spline. An instrument portion is releasably coupled to the stage portion and includes one or more additional layers of the carriage removably coupled to the first layer, and an elongate shaft extending distally from the one or more additional layers and having an end effector arranged at a distal end of the elongate shaft. The elongate shaft and the end effector penetrate the elevator layer and the first end when the instrument portion is coupled to the stage portion.

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Background/Summary

TECHNICAL FIELD

[0001] The systems and methods disclosed herein are directed to surgical tools and, more particularly to, a robotic surgical tool having a stage portion and an instrument portion removably secured to the stage portion.

BACKGROUND

[0002] Minimally invasive surgical (MIS) instruments are often preferred over traditional open surgical devices due to the reduced post-operative recovery time and minimal scarring. The most common MIS procedure may be endoscopy, and the most common form of endoscopy is laparoscopy, in which one or more small incisions are formed in the abdomen of a patient and a trocar is inserted through the incision to form a pathway that provides access to the abdominal cavity. The cannula and sealing system of the trocar is used to introduce various instruments and tools into the abdominal cavity, as well as to provide insufflation to elevate the abdominal wall above the organs. The instruments can be used to engage and/or treat tissue in a number of ways to achieve a diagnostic or therapeutic effect.

[0003] Each surgical tool typically includes an end effector arranged at its distal end. Example end effectors include clamps, graspers, scissors, staplers, suction irrigators, blades (i.e., RF), and needle holders, and are similar to those used in conventional (open) surgery except that the end effector of each tool is separated from its handle by an approximately 12-inch long shaft. A camera or image capture device, such as an endoscope, is also commonly introduced into the abdominal cavity to enable the surgeon to view the surgical field and the operation of the end effectors during operation. The surgeon is able to view the procedure in real-time by means of a visual display in communication with the image capture device.

[0004] Various robotic systems have recently been developed to assist in MIS procedures. Robotic systems can allow for more intuitive hand movements by maintaining natural eye-hand axis. Robotic systems can also allow for more degrees of freedom in movement by including a “wrist” joint that creates a more natural hand-like articulation and allows for access to hard to reach spaces. The instrument's end effector can be articulated (moved) using motors and actuators forming part of a computerized motion system. A user (e.g., a surgeon) is able to remotely operate an instrument's end effector by grasping and manipulating in space one or more controllers that communicate with an instrument driver coupled to the surgical instrument. User inputs are processed by a computer system incorporated into the robotic surgical system and the instrument driver responds by actuating the motors and actuators of the motion system. Moving the drive cables and/or other mechanical mechanisms manipulates the end effector to desired positions and configurations.

[0005] Improvements to robotically-enabled medical systems will provide physicians with the ability to perform endoscopic and laparoscopic procedures more effectively and with improved ease.

SUMMARY OF DISCLOSURE

[0006] Various details of the present disclosure are hereinafter summarized to provide a basic understanding. This summary is not an extensive overview of the disclosure and is neither intended to identify certain elements of the disclosure, nor to delineate the scope thereof. Rather, the primary purpose of this summary is to present some concepts of the disclosure in a simplified form prior to the more detailed description that is presented hereinafter.

[0007] Embodiments disclosed herein include a robotic surgical tool that includes a stage assembly and a core assembly removably mountable to the stage assembly. The stage assembly includes a lead screw and at least one spline extendable between first and second ends of the stage assembly, and a nut rotatably mounted to the lead screw to translate between the first and second ends upon rotation of the lead screw. The core assembly includes a drive housing that is movably mountable to the stage assembly to translate between the first and second end with the nut. In a further embodiment, the stage assembly further comprises a carriage within which the nut is provided such that the carriage translates axially between the first and second ends with the nut upon rotation of the lead screw, wherein the drive housing is removably coupled to the carriage to translate with the carriage. In another further embodiment, the carriage defines a trough that receives the drive housing to removably couple the drive housing to the carriage. In another further embodiment, the trough is defined by a pair of sidewalls of the carriage that engage and retain the drive housing within the trough. In another further embodiment, the surgical tool further comprises a drive gear coupled to the at least one spline and rotatable with rotation of the at least one spline, wherein the drive gear is slidable on the at least one spline, and an activating mechanism provided in the drive housing and operatively coupled to the drive gear such that rotation of the drive gear correspondingly actuates the activating mechanism. In another further embodiment, the nut is provided in a carriage that translates axially between the first and second end with the nut upon rotation of the lead screw, wherein the drive gear is housed on the carriage and, upon removably mounting the drive housing to the carriage, the drive gear is operatively coupled to the activating mechanism such that rotation of the drive gear correspondingly actuates the activating mechanism. In another further embodiment, the at least one spline extends through an associated spline channel formed in the carriage. In another further embodiment, the drive housing is removably coupled within a trough of the carriage. In another further embodiment, at least one tooth of the drive gear extends into the trough to interact with the activating mechanism when the drive housing is removably mounted in the trough of the carriage. In another further embodiment, the surgical tool further comprises a drive input arranged at the first end and operatively coupled to the at least one spline such that rotation of the drive input correspondingly rotates the at least one spline and the drive gear, and an instrument driver arranged at an end of a robotic arm and matable with the stage assembly at the first end, the instrument driver providing a drive output matable with the drive input such that rotation of the drive output correspondingly rotates the drive input and thereby actuates the activating mechanism. In another further embodiment, the stage assembly further comprises a pair of guide rails extendable between the first and second ends, and a pair of laterally extending arms are provided on the drive housing for removably and slidingly attaching the drive housing to the pair of guide rails. In another further embodiment, the surgical tool further comprises a primary shroud at least partially enclosing the core assembly when installed on the stage assembly, and a secondary shroud pivotably attached to the primary shroud at a hinge, wherein the secondary shroud is movable between a closed position, where the secondary shroud occludes an opening in the primary shroud, and an open position, where the opening in the primary shroud is exposed. In another further embodiment, the surgical tool further comprises a primary shroud at least partially enclosing the core assembly when installed on the stage assembly, and a secondary shroud slidingly attached to the primary shroud, wherein the secondary shroud is circumferentially revolvable relative to the primary shroud between a closed position, where the secondary shroud occludes an opening in the primary shroud, and an open position, where the opening in the primary shroud is exposed. In another further embodiment, the surgical tool further comprises a primary shroud at least partially enclosing the core assembly when installed on the stage assembly, and a secondary shroud slidingly attached to the primary shroud, wherein the secondary shroud is axially slidable relative to the primary shroud between a closed position, where the secondary shroud occludes an opening in the primary shroud, and an open position, where the opening in the primary shroud is exposed. In another further embodiment, the core assembly further comprises an elongate shaft extending distally from the drive housing and penetrating the first end of the stage assembly when the drive housing is coupled to the stage assembly, and an end effector arranged at a distal end of the elongate shaft.

[0008] Embodiments disclosed herein may further include a robotic surgical tool that includes a stage assembly having a lead screw and at least one spline extendable between first and second ends of the stage assembly, a carriage slidingly disposed on the at least one spline, and a carriage nut attached to the carriage and rotatably mounted to the lead screw to translate the carriage between the first and second ends upon rotation of the lead screw. A core assembly having a drive housing is removably mountable to the carriage, wherein the core assembly is axially translatable when the drive housing is installed on the carriage. A shroud assembly is provided having a primary shroud and a secondary shroud, where the primary shroud at least partially encloses the core assembly when installed on the stage assembly, and the secondary shroud being movable between a first position, where the secondary shroud occludes an opening in the primary shroud, and a second position, where the opening in the primary shroud is exposed. In a further embodiment, the secondary shroud is pivotally attached to the primary shroud at a hinge. In another further embodiment, the secondary shroud is slidingly attached to the primary shroud, and the secondary shroud is circumferentially revolvable relative to the primary shroud between the first and second positions. In another further embodiment, the secondary shroud is slidingly attached to the primary shroud, with the secondary shroud being axially slidable relative to the primary shroud between the first and second positions.

[0009] Embodiments disclosed herein may further include a robotic surgical tool that includes a stage assembly, a core assembly, and a shroud assembly. The stage assembly includes a lead screw and at least one spline extendable between first and second ends of the stage assembly, a carriage slidingly disposed on the at least one spline, a carriage nut attached to the carriage and rotatably mounted to the lead screw to translate the carriage between the first and second ends upon rotation of the lead screw, and a drive gear provided in the carriage and coupled to the at least one spline to be rotatable with rotation of the at least one spline and axially slidable on the at least one spline with translation of the carriage. The core assembly includes a drive housing removably mountable to the carriage, an activating mechanism arranged within the drive housing, an elongate shaft operatively coupled to the activating mechanism and extending distally from the drive housing and penetrating the first end of the stage assembly when the drive housing is coupled to the carriage, an end effector arranged at a distal end of the elongate shaft and movable via actuation of the activating mechanism, wherein, when the drive housing is installed on the carriage, the core assembly is axially translatable and the activating mechanism is operatively coupled to the drive gear such that rotation of the drive gear correspondingly actuates the activating mechanism. The shroud assembly includes a primary shroud and a secondary shroud, the primary shroud at least partially enclosing the core assembly when installed on the stage assembly, and the secondary shroud being movable between a first position, where the secondary shroud occludes an opening in the primary shroud, and a second position, where the opening in the primary shroud is exposed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The disclosed aspects will hereinafter be described in conjunction with the appended drawings, provided to illustrate and not to limit the disclosed aspects, wherein like designations denote like elements.

[0011] FIG. **1** illustrates an embodiment of a cart-based robotic system arranged for diagnostic and/or therapeutic bronchoscopy procedure(s).

[0012] FIG. **2** depicts further aspects of the robotic system of FIG. **1**.

[0013] FIG. **3**A illustrates an embodiment of the robotic system of FIG. **1** arranged for ureteroscopy.

[0014] FIG. **3**B illustrates an embodiment of the robotic system of FIG. **1** arranged for a vascular procedure.

[0015] FIG. **4** illustrates an embodiment of a table-based robotic system arranged for a bronchoscopy procedure.

[0016] FIG. **5** provides an alternative view of the robotic system of FIG. **4**.

[0017] FIG. **6** illustrates an example system configured to stow robotic arm(s).

[0018] FIG. **7**A illustrates an embodiment of a table-based robotic system configured for a ureteroscopy procedure.

[0019] FIG. **7**B illustrates an embodiment of a table-based robotic system configured for a laparoscopic procedure.

[0020] FIG. **7**C illustrates an embodiment of the table-based robotic system of FIGS. **4**-**7**B with pitch or tilt adjustment.

[0021] FIG. **8** provides a detailed illustration of the interface between the table and the column of the table-based robotic system of FIGS. **4**-**7**.

[0022] FIG. **9**A illustrates an alternative embodiment of a table-based robotic system.

[0023] FIG. **9**B illustrates an end view of the table-based robotic system of FIG. **9**A.

[0024] FIG. **9**C illustrates an end view of a table-based robotic system with robotic arms attached thereto.

[0025] FIG. **10** illustrates an exemplary instrument driver.

[0026] FIG. **11** illustrates an exemplary medical instrument with a paired instrument driver.

[0027] FIG. **12** illustrates an alternative design for an instrument driver and instrument where the axes of the drive units are parallel to the axis of the elongated shaft of the instrument.

[0028] FIG. **13** illustrates an instrument having an instrument-based insertion architecture.

[0029] FIG. **14** illustrates an exemplary controller.

[0030] FIG. **15** depicts a block diagram illustrating a localization system that estimates a location of one or more elements of the robotic systems of FIGS. **1**-**7**C, such as the location of the instrument of FIGS. **11**-**13**, in accordance to an example embodiment.

[0031] FIG. **16** is an isometric side view of an example surgical tool that may incorporate some or all of the principles of the present disclosure.

[0032] FIG. **17** is an isometric view of the surgical tool of FIG. **16** when unassembled from its shroud assembly, according to one or more embodiments.

[0033] FIG. **18**A is an isometric view of the tool of FIGS. **16**-**17** releasably coupled to an example instrument driver, according to one or more embodiments.

[0034] FIG. **18**B provides separated isometric end views of the instrument driver of FIG. **18**A and the surgical tool of FIGS. **16**-**17**.

[0035] FIG. **19** illustrates the surgical tool of FIGS. **16**-**17** having a stage portion and an instrument portion when removed from a stage portion, according to one or more embodiments.

[0036] FIG. **20**A provides an isometric end view of a portion of the instrument portion of FIG. **19**.

[0037] FIG. **20**B provides an isometric end view of a portion of the stage portion of FIG. **19**.
[0038] FIG. **21** illustrates a handle of the instrument portion depicted in FIG. **20**A when unassembled from an elevator of the stage portion depicted in FIG. **20**B and when unassembled from a table alignment tool, according to one or more embodiments.
[0039] FIG. **22** illustrates table alignment tool of FIG. **21** being utilized to align the instrument portion and the stage portion of the surgical tool such that the handle of the instrument portion may be releasably secured to the elevator of the stage portion, according to one or more embodiments.
[0040] FIGS. **23**A-**23**I illustrate example methods of operating and assembling a surgical tool, according to one or more embodiments.
[0041] FIGS. **24**A-**24**B illustrate respective bottom end, top end, and partially-exploded views of the removable cap of FIG. **19**, according to one or more embodiments.
[0042] FIGS. **25**A-**25**B illustrate splines configured to telescope from a handle interface when in a nested condition and when in an at least partially unnested condition, respectively, according to one or more embodiments of the present disclosure.
[0043] FIGS. **26**A-**26**B illustrate splines configured to telescope from an elevator of the carriage when in a nested condition and when in an at least partially unnested condition, respectively, according to one or more embodiments of the present disclosure.
[0044] FIG. **27** illustrates a telescoping spline configuration, according to one or more embodiments.
[0045] FIGS. **28**A-**28**B illustrate a shroud configured to expand and contract with the telescoping splines in respective collapsed and expanded positions, according to one or more embodiments.
[0046] FIG. **29** is an isometric side view of an example stage for a surgical tool utilizing a translatable drive puck configured to map spline input to different output locations, according to one or more embodiments.
[0047] FIGS. **30**A and **30**B illustrate a proximal face and a distal face, respectively, of the translatable drive puck of FIG. **29** when unassembled.
[0048] FIG. **31** illustrates internal drive components and gearing arrangeable within the translatable drive puck of FIG. **29**, according to one or more embodiments.
[0049] FIG. **32** illustrates a removable handle segment of the surgical tool adapted to be releasably secured on the translatable drive puck, according to one or more embodiments.
[0050] FIG. **33** is a close up of the removable handle segment of FIG. **32** illustrating exemplary internal drive mechanisms, according to one or more embodiments.
[0051] FIG. **34** illustrates how the removable portion may be aligned with the drive puck during installation.
[0052] FIG. **35** illustrates the surgical tool wherein the removable handle segment of FIG. **32** has been installed within the stage of FIG. **29**.
[0053] FIG. **36** illustrates the surgical tool having a stage sub-assembly on which a modular handle sub-assembly may be installed, according to various embodiments.
[0054] FIG. **37** illustrates an exemplary installation of the handle sub-assembly on the stage sub-assembly shown in FIG. **36**.
[0055] FIG. **38** illustrates a core assembly of the surgical tool, according to various other embodiments.
[0056] FIG. **39** illustrates an exemplary stage assembly on which the core assembly of FIG. **38** may be mounted.
[0057] FIG. **40** illustrates the surgical tool incorporating yet another alternate handle sub-assembly configured to be dropped on or angled into a stage sub-assembly, according to various embodiments.
[0058] FIG. **41** is a cross-sectional side view of a portion of the handle sub-assembly and the stage sub-assembly shown in FIG. **40**.
[0059] FIGS. **42**A-**42**D illustrate various means of providing access through which a handle sub-

assembly may be dropped on or angled into a stage sub-assembly, according to various embodiments.

DETAILED DESCRIPTION

1. Overview

[0060] Aspects of the present disclosure may be integrated into a robotically-enabled medical system capable of performing a variety of medical procedures, including both minimally invasive (e.g., laparoscopy) and non-invasive (e.g., endoscopy) procedures. Among endoscopy procedures, the system may be capable of performing bronchoscopy, ureteroscopy, gastroscopy, etc.

[0061] In addition to performing the breadth of procedures, the system may provide additional benefits, such as enhanced imaging and guidance to assist the physician. Additionally, the system may provide the physician with the ability to perform the procedure from an ergonomic position without the need for awkward arm motions and positions. Still further, the system may provide the physician with the ability to perform the procedure with improved ease of use such that one or more of the instruments of the system can be controlled by a single user.

[0062] Various embodiments will be described below in conjunction with the drawings for purposes of illustration. It should be appreciated that many other implementations of the disclosed concepts are possible, and various advantages can be achieved with the disclosed implementations. Headings are included herein for reference and to aid in locating various sections. These headings are not intended to limit the scope of the concepts described with respect thereto. Such concepts may have applicability throughout the entire specification.

A. Robotic System—Cart

[0063] The robotically-enabled medical system may be configured in a variety of ways depending on the particular procedure. FIG. **1** illustrates an embodiment of a cart-based robotically-enabled system **100** arranged for a diagnostic and/or therapeutic bronchoscopy procedure. For a bronchoscopy procedure, the robotic system **100** may include a cart **102** having one or more robotic arms **104** (three shown) to deliver a medical instrument (alternately referred to as a “surgical tool”), such as a steerable endoscope **106** (e.g., a procedure-specific bronchoscope for bronchoscopy), to a natural orifice access point (i.e., the mouth of the patient) to deliver diagnostic and/or therapeutic tools. As shown, the cart **102** may be positioned proximate to the patient's upper torso in order to provide access to the access point. Similarly, the robotic arms **104** may be actuated to position the bronchoscope relative to the access point. The arrangement in FIG. **1** may also be utilized when performing a gastro-intestinal (GI) procedure with a gastroscope, a specialized endoscope for GI procedures.

[0064] Once the cart **102** is properly positioned adjacent the patient, the robotic arms **104** are operated to insert the steerable endoscope **106** into the patient robotically, manually, or a combination thereof. The steerable endoscope **106** may comprise at least two telescoping parts, such as an inner leader portion and an outer sheath portion, where each portion is coupled to a separate instrument driver of a set of instrument drivers **108** (alternately referred to as “tool drivers”). As illustrated, each instrument driver **108** is coupled to the distal end of a corresponding one of the robotic arms **104**. This linear arrangement of the instrument drivers **108**, which facilitates coaxially aligning the leader portion with the sheath portion, creates a “virtual rail” **110** that may be repositioned in space by manipulating the robotic arms **104** into different angles and/or positions. Translation of the instrument drivers **108** along the virtual rail **110** telescopes the inner leader portion relative to the outer sheath portion, thus effectively advancing or retracting the endoscope **106** relative to the patient.

[0065] As illustrated, the virtual rail **110** (and other virtual rails described herein) is depicted in the drawings using dashed lines, thus not constituting any physical structure of the system **100**. The angle of the virtual rail **110** may be adjusted, translated, and pivoted based on clinical application or physician preference. For example, in bronchoscopy, the angle and position of the virtual rail **110** as shown represents a compromise between providing physician access to the endoscope **106** while

minimizing friction that results from bending the endoscope **106** into the patient's mouth.
[0066] After insertion into the patient's mouth, the endoscope **106** may be directed down the patient's trachea and lungs using precise commands from the robotic system **100** until reaching a target destination or operative site. In order to enhance navigation through the patient's lung network and/or reach the desired target, the endoscope **106** may be manipulated to telescopically extend the inner leader portion from the outer sheath portion to obtain enhanced articulation and greater bend radius. The use of separate instrument drivers **108** also allows the leader portion and sheath portion to be driven independent of each other.
[0067] For example, the endoscope **106** may be directed to deliver a biopsy needle to a target, such as, for example, a lesion or nodule within the lungs of a patient. The needle may be deployed down a working channel that runs the length of the endoscope **106** to obtain a tissue sample to be analyzed by a pathologist. Depending on the pathology results, additional tools may be deployed down the working channel of the endoscope for additional biopsies. After identifying a tissue sample to be malignant, the endoscope **106** may endoscopically deliver tools to resect the potentially cancerous tissue. In some instances, diagnostic and therapeutic treatments can be delivered in separate procedures. In those circumstances, the endoscope **106** may also be used to deliver a fiducial marker to “mark” the location of a target nodule as well. In other instances, diagnostic and therapeutic treatments may be delivered during the same procedure.
[0068] The system **100** may also include a movable tower **112**, which may be connected via support cables to the cart **102** to provide support for controls, electronics, fluidics, optics, sensors, and/or power to the cart **102**. Placing such functionality in the tower **112** allows for a smaller form factor cart **102** that may be more easily adjusted and/or re-positioned by an operating physician and his/her staff. Additionally, the division of functionality between the cart/table and the support tower **112** reduces operating room clutter and facilitates improving clinical workflow. While the cart **102** may be positioned close to the patient, the tower **112** may alternatively be stowed in a remote location to stay out of the way during a procedure.
[0069] In support of the robotic systems described above, the tower **112** may include component(s) of a computer-based control system that stores computer program instructions, for example, within a non-transitory computer-readable storage medium such as a persistent magnetic storage drive, solid state drive, etc. The execution of those instructions, whether the execution occurs in the tower **112** or the cart **102**, may control the entire system or sub-system(s) thereof. For example, when executed by a processor of the computer system, the instructions may cause the components of the robotics system to actuate the relevant carriages and arm mounts, actuate the robotics arms, and control the medical instruments. For example, in response to receiving the control signal, motors in the joints of the robotic arms **104** may position the arms into a certain posture or angular orientation.
[0070] The tower **112** may also include one or more of a pump, flow meter, valve control, and/or fluid access in order to provide controlled irrigation and aspiration capabilities to the system **100** that may be deployed through the endoscope **106**. These components may also be controlled using the computer system of the tower **112**. In some embodiments, irrigation and aspiration capabilities may be delivered directly to the endoscope **106** through separate cable(s).
[0071] The tower **112** may include a voltage and surge protector designed to provide filtered and protected electrical power to the cart **102**, thereby avoiding placement of a power transformer and other auxiliary power components in the cart **102**, resulting in a smaller, more moveable cart **102**.
[0072] The tower **112** may also include support equipment for sensors deployed throughout the robotic system **100**. For example, the tower **112** may include opto-electronics equipment for detecting, receiving, and processing data received from optical sensors or cameras throughout the robotic system **100**. In combination with the control system, such opto-electronics equipment may be used to generate real-time images for display in any number of consoles deployed throughout the system, including in the tower **112**. Similarly, the tower **112** may also include an electronic

subsystem for receiving and processing signals received from deployed electromagnetic (EM) sensors. The tower **112** may also be used to house and position an EM field generator for detection by EM sensors in or on the medical instrument.

[0073] The tower **112** may also include a console **114** in addition to other consoles available in the rest of the system, e.g., a console mounted to the cart **102**. The console **114** may include a user interface and a display screen (e.g., a touchscreen) for the physician operator. Consoles in the system **100** are generally designed to provide both robotic controls as well as pre-operative and real-time information of the procedure, such as navigational and localization information of a medical tool. When the console **114** is not the only console available to the physician, it may be used by a second operator, such as a nurse, to monitor the health or vitals of the patient and the operation of system, as well as provide procedure-specific data, such as navigational and localization information. In other embodiments, the console **114** may be housed in a body separate from the tower **112**.

[0074] The tower **112** may be coupled to the cart **102** and endoscope **106** through one or more cables **116** connections. In some embodiments, support functionality from the tower **112** may be provided through a single cable **116** extending to the cart **102**, thus simplifying and de-cluttering the operating room. In other embodiments, specific functionality may be coupled in separate cabling and connections. For example, while power may be provided through a single power cable to the cart **102**, support for controls, optics, fluidics, and/or navigation may be provided through one or more separate cables.

[0075] FIG. **2** provides a detailed illustration of an embodiment of the cart **102** from the cart-based robotically-enabled system **100** of FIG. **1**. The cart **102** generally includes an elongated support structure **202** (also referred to as a “column”), a cart base **204**, and a console **206** at the top of the column **202**. The column **202** may include one or more carriages, such as a carriage **208** (alternatively “arm support”) for supporting the deployment of the robotic arms **104**. The carriage **208** may include individually configurable arm mounts that rotate along a perpendicular axis to adjust the base **214** of the robotic arms **104** for better positioning relative to the patient. The carriage **208** also includes a carriage interface **210** that allows the carriage **208** to vertically translate along the column **202**.

[0076] The carriage interface **210** is connected to the column **202** through slots, such as slot **212**, that are positioned on opposite sides of the column **202** to guide the vertical translation of the carriage **208**. The slot **212** contains a vertical translation interface to position and hold the carriage **208** at various vertical heights relative to the cart base **204**. Vertical translation of the carriage **208** allows the cart **102** to adjust the reach of the robotic arms **104** to meet a variety of table heights, patient sizes, and physician preferences. Similarly, the individually configurable arm mounts on the carriage **208** allow a base **214** of the robotic arms **104** to be angled in a variety of configurations.

[0077] In some embodiments, the slot **212** may be supplemented with slot covers (not shown) that are flush and parallel to the slot surface to prevent dirt and fluid ingress into the internal chambers of the column **202** and the vertical translation interface as the carriage **208** vertically translates. The slot covers may be deployed through pairs of spring spools positioned near the vertical top and bottom of the slot **212**. The covers are coiled within the spools until deployed to extend and retract from their coiled state as the carriage **208** vertically translates up and down. The spring-loading of the spools provides force to retract the cover into a spool when carriage **208** translates towards the spool, while also maintaining a tight seal when the carriage **208** translates away from the spool. The covers may be connected to the carriage **208** using, for example, brackets in the carriage interface **210** to ensure proper extension and retraction of the cover as the carriage **208** translates.

[0078] The column **202** may internally comprise mechanisms, such as gears and motors, that are designed to use a vertically aligned lead screw to translate the carriage **208** in a mechanized fashion in response to control signals generated in response to user inputs, e.g., inputs from the console **206**.

[0079] The robotic arms **104** may generally comprise robotic arm bases **214** and end effectors **216** (three shown), separated by a series of linkages **218** connected by a corresponding series of joints **220**, each joint **220** including an independent actuator, and each actuator including an independently controllable motor. Each independently controllable joint **220** represents an independent degree of freedom available to the corresponding robotic arm **104**. In the illustrated embodiment, each arm **104** has seven joints **220**, providing seven degrees of freedom. A multitude of joints **220** result in a multitude of degrees of freedom, allowing for “redundant” degrees of freedom. Redundant degrees of freedom allow the robotic arms **104** to position its respective end effectors **216** at a specific position, orientation, and trajectory in space using different linkage positions and joint angles. This allows for the system **100** to position and direct a medical instrument from a desired point in space while allowing the physician to move the arm joints **220** into a clinically advantageous position away from the patient to create greater access, while avoiding arm collisions.
[0080] The cart base **204** balances the weight of the column **202**, carriage **208**, and arms **104** over the floor. Accordingly, the cart base **204** houses heavier components, such as electronics, motors, power supply, as well as components that either enable movement and/or immobilize the cart. For example, the cart base **204** includes rollable casters **222** that allow for the cart **102** to easily move around a room prior to a procedure. After reaching an appropriate position, the casters **222** may be immobilized using locks to hold the cart **102** in place during the procedure.
[0081] Positioned at the vertical end of the column **202**, the console **206** allows for both a user interface for receiving user input and a display screen (or a dual-purpose device such as, for example, a touchscreen **224**) to provide the physician user with both pre-operative and intra-operative data. Potential pre-operative data on the touchscreen **224** may include pre-operative plans, navigation and mapping data derived from pre-operative computerized tomography (CT) scans, and/or notes from pre-operative patient interviews. Intra-operative data on the touchscreen **224** may include optical information provided from the tool, sensor and coordinate information from sensors, as well as vital patient statistics, such as respiration, heart rate, and/or pulse. The console **206** may be positioned and tilted to allow a physician to access the console from the side of the column **202** opposite carriage **208**. From this position, the physician may view the console **206**, the robotic arms **104**, and the patient while operating the console **206** from behind the cart **102**. As shown, the console **206** also includes a handle **226** to assist with maneuvering and stabilizing cart **102**.
[0082] FIG. **3**A illustrates an embodiment of the system **100** of FIG. **1** arranged for ureteroscopy. In a ureteroscopic procedure, the cart **102** may be positioned to deliver a ureteroscope **302**, a procedure-specific endoscope designed to traverse a patient's urethra and ureter, to the lower abdominal area of the patient. In ureteroscopy, it may be desirable for the ureteroscope **302** to be directly aligned with the patient's urethra to reduce friction and forces on the sensitive anatomy. As shown, the cart **102** may be aligned at the foot of the table to allow the robotic arms **104** to position the ureteroscope **302** for direct linear access to the patient's urethra. From the foot of the table, the robotic arms **104** may insert the ureteroscope **302** along a virtual rail **304** directly into the patient's lower abdomen through the urethra.
[0083] After insertion into the urethra, using similar control techniques as in bronchoscopy, the ureteroscope **302** may be navigated into the bladder, ureters, and/or kidneys for diagnostic and/or therapeutic applications. For example, the ureteroscope **302** may be directed into the ureter and kidneys to break up kidney stone build-up using a laser or ultrasonic lithotripsy device deployed down a working channel of the ureteroscope **302**. After lithotripsy is complete, the resulting stone fragments may be removed using baskets deployed down the working channel of the ureteroscope **302**.
[0084] FIG. **3**B illustrates another embodiment of the system **100** of FIG. **1** arranged for a vascular procedure. In a vascular procedure, the system **100** may be configured such that the cart **102** may

deliver a medical instrument **306**, such as a steerable catheter, to an access point in the femoral artery in the patient's leg. The femoral artery presents both a larger diameter for navigation as well as a relatively less circuitous and tortuous path to the patient's heart, which simplifies navigation. As in an ureteroscopic procedure, the cart **102** may be positioned towards the patient's legs and lower abdomen to allow the robotic arms **104** to provide a virtual rail **308** with direct linear access to the femoral artery access point in the patient's thigh/hip region. After insertion into the artery, the medical instrument **306** may be directed and advanced by translating the instrument drivers **108**. Alternatively, the cart **102** may be positioned around the patient's upper abdomen in order to reach alternative vascular access points, such as, for example, the carotid and brachial arteries near the patient's shoulder and wrist.

B. Robotic System—Table

[0085] Embodiments of the robotically-enabled medical system may also incorporate the patient's table. Incorporation of the table reduces the amount of capital equipment within the operating room by removing the cart, which allows greater access to the patient. FIG. **4** illustrates an embodiment of such a robotically-enabled system **400** arranged for a bronchoscopy procedure. As illustrated, the system **400** includes a support structure or column **402** for supporting platform **404** (shown as a “table” or “bed”) over the floor. Much like in the cart-based systems **100**, end effectors of the robotic arms **406** of the system **400** comprise instrument drivers **408** (alternately referred to as “tool drivers”) that are designed to manipulate an elongated medical instrument, such as a bronchoscope **410**, through or along a virtual rail **412** formed from the linear alignment of the instrument drivers **408**. In practice, a

[0086] C-arm for providing fluoroscopic imaging may be positioned over the patient's upper abdominal area by placing the emitter and detector around the table **404**.

[0087] FIG. **5** provides an alternative view of the system **400** without the patient and medical instrument for discussion purposes. As shown, the column **402** may include one or more carriages **502** shown as ring-shaped in the system **400**, from which the one or more robotic arms **406** may be based. The carriages **502** may translate along a vertical column interface **504** that runs the length (height) of the column **402** to provide different vantage points from which the robotic arms **406** may be positioned to reach the patient. The carriage(s) **502** may rotate around the column **402** using a mechanical motor positioned within the column **402** to allow the robotic arms **406** to have access to multiples sides of the table **404**, such as, for example, both sides of the patient. In embodiments with multiple carriages **502**, the carriages **502** may be individually positioned on the column **402** and may translate and/or rotate independent of the other carriages **502**. While carriages **502** need not surround the column **402** or even be circular, the ring-shape as shown facilitates rotation of the carriages **502** around the column **402** while maintaining structural balance. Rotation and translation of the carriages **502** allows the system **400** to align medical instruments, such as endoscopes and laparoscopes, into different access points on the patient.

[0088] In other embodiments (discussed in greater detail below with respect to FIG. **9**A), the system **400** can include a patient table or bed with adjustable arm supports in the form of bars or rails extending alongside it. One or more robotic arms **406** (e.g., via a shoulder with an elbow joint) can be attached to the adjustable arm supports, which can be vertically adjusted. By providing vertical adjustment, the robotic arms **406** are advantageously capable of being stowed compactly beneath the patient table or bed, and subsequently raised during a procedure.

[0089] The arms **406** may be mounted on the carriages **502** through a set of arm mounts **506** comprising a series of joints that may individually rotate and/or telescopically extend to provide additional configurability to the robotic arms **406**. Additionally, the arm mounts **506** may be positioned on the carriages **502** such that when the carriages **502** are appropriately rotated, the arm mounts **506** may be positioned on either the same side of the table **404** (as shown in FIG. **5**), on opposite sides of table **404** (as shown in FIG. **7**B), or on adjacent sides of the table **404** (not shown).

[0090] The column **402** structurally provides support for the table **404**, and a path for vertical translation of the carriages **502**. Internally, the column **402** may be equipped with lead screws for guiding vertical translation of the carriages, and motors to mechanize the translation of said carriages based the lead screws. The column **402** may also convey power and control signals to the carriage **502** and robotic arms **406** mounted thereon.
[0091] A table base **508** serves a similar function as the cart base **204** of the cart **102** shown in FIG. **2**, housing heavier components to balance the table/bed **404**, the column **402**, the carriages **502**, and the robotic arms **406**. The table base **508** may also incorporate rigid casters to provide stability during procedures. Deployed from the bottom of the table base **508**, the casters may extend in opposite directions on both sides of the base **508** and retract when the system **400** needs to be moved.
[0092] In some embodiments, the system **400** may also include a tower (not shown) that divides the functionality of system **400** between table and tower to reduce the form factor and bulk of the table **404**. As in earlier disclosed embodiments, the tower may provide a variety of support functionalities to the table **404**, such as processing, computing, and control capabilities, power, fluidics, and/or optical and sensor processing. The tower may also be movable to be positioned away from the patient to improve physician access and de-clutter the operating room. Additionally, placing components in the tower allows for more storage space in the table base **508** for potential stowage of the robotic arms **406**. The tower may also include a master controller or console that provides both a user interface for user input, such as keyboard and/or pendant, as well as a display screen (or touchscreen) for pre-operative and intra-operative information, such as real-time imaging, navigation, and tracking information. In some embodiments, the tower may also contain holders for gas tanks to be used for insufflation.
[0093] In some embodiments, a table base may stow and store the robotic arms when not in use. FIG. **6** illustrates an embodiment of the system **400** that is configured to stow robotic arms **606** within a table base **406**. In the system **400**, one or more carriages **602** (one shown) may be vertically translated into a base **604** to stow one or more robotic arms **606**, one or more arm mounts **608**, and the carriages **602** within the base **604**. Base covers **610** may be translated and retracted open to deploy the carriages **602**, the arm mounts **608**, and the arms **606** around the column **612**, and closed to stow and protect them when not in use. The base covers **610** may be sealed with a membrane **614** along the edges of its opening to prevent dirt and fluid ingress when closed.
[0094] FIG. **7**A illustrates an embodiment of the robotically-enabled table-based system **400** configured for a ureteroscopy procedure. In ureteroscopy, the table **404** may include a swivel portion **702** for positioning a patient off-angle from the column **402** and the table base **508**. The swivel portion **702** may rotate or pivot around a pivot point (e.g., located below the patient's head) in order to position the bottom portion of the swivel portion **702** away from the column **402**. For example, the pivoting of the swivel portion **702** allows a C-arm (not shown) to be positioned over the patient's lower abdomen without competing for space with the column (not shown) below table **404**. By rotating the carriage (not shown) around the column **402**, the robotic arms **406** may directly insert a ureteroscope **704** along a virtual rail **706** into the patient's groin area to reach the urethra. In ureteroscopy, stirrups **708** may also be fixed to the swivel portion **702** of the table **404** to support the position of the patient's legs during the procedure and allow clear access to the patient's groin area.
[0095] FIG. **7**B illustrates an embodiment of the system **400** configured for a laparoscopic procedure. In a laparoscopic procedure, through small incision(s) in the patient's abdominal wall, minimally invasive instruments may be inserted into the patient's anatomy. In some embodiments, the minimally invasive instruments comprise an elongated rigid member, such as a shaft, which is used to access anatomy within the patient. After inflation of the patient's abdominal cavity, the instruments may be directed to perform surgical or medical tasks, such as grasping, cutting, ablating, suturing, etc. In some embodiments, the instruments can comprise a scope, such as a

laparoscope. As shown in FIG. **7**B, the carriages **502** of the system **400** may be rotated and vertically adjusted to position pairs of the robotic arms **406** on opposite sides of the table **404**, such that an instrument **710** may be positioned using the arm mounts **506** to be passed through minimal incisions on both sides of the patient to reach his/her abdominal cavity.

[0096] To accommodate laparoscopic procedures, the system **400** may also tilt the platform to a desired angle. FIG. **7**C illustrates an embodiment of the system **400** with pitch or tilt adjustment. As shown in FIG. **7**C, the system **400** may accommodate tilt of the table **404** to position one portion of the table **404** at a greater distance from the floor than the other. Additionally, the arm mounts **506** may rotate to match the tilt such that the arms **406** maintain the same planar relationship with table **404**. To accommodate steeper angles, the column **402** may also include telescoping portions **712** that allow vertical extension of the column **402** to keep the table **404** from touching the floor or colliding with the base **508**.

[0097] FIG. **8** provides a detailed illustration of the interface between the table **404** and the column **402**. Pitch rotation mechanism **802** may be configured to alter the pitch angle of the table **404** relative to the column **402** in multiple degrees of freedom. The pitch rotation mechanism **802** may be enabled by the positioning of orthogonal axes A and B at the column-table interface, each axis actuated by a separate motor **804***a* and **804***b* responsive to an electrical pitch angle command. Rotation along one screw **806***a* would enable tilt adjustments in one axis A, while rotation along another screw **806***b* would enable tilt adjustments along the other axis B. In some embodiments, a ball joint can be used to alter the pitch angle of the table **404** relative to the column **402** in multiple degrees of freedom.

[0098] For example, pitch adjustments are particularly useful when trying to position the table in a Trendelenburg position, i.e., position the patient's lower abdomen at a higher position from the floor than the patient's lower abdomen, for lower abdominal surgery. The Trendelenburg position causes the patient's internal organs to slide towards his/her upper abdomen through the force of gravity, clearing out the abdominal cavity for minimally invasive tools to enter and perform lower abdominal surgical or medical procedures, such as laparoscopic prostatectomy.

[0099] FIGS. **9**A and **9**B illustrate isometric and end views, respectively, of an alternative embodiment of a table-based surgical robotics system **900**. The surgical robotics system **900** includes one or more adjustable arm supports **902** that can be configured to support one or more robotic arms (see, for example, FIG. **9**C) relative to a table **904**. In the illustrated embodiment, a single adjustable arm support **902** is shown, though an additional arm support can be provided on an opposite side of the table **904**. The adjustable arm support **902** can be configured so that it can move relative to the table **904** to adjust and/or vary the position of the adjustable arm support **902** and/or any robotic arms mounted thereto relative to the table **904**. For example, the adjustable arm support **902** may be adjusted in one or more degrees of freedom relative to the table **904**. The adjustable arm support **902** provides high versatility to the system **900**, including the ability to easily stow the one or more adjustable arm supports **902** and any robotics arms attached thereto beneath the table **904**. The adjustable arm support **902** can be elevated from the stowed position to a position below an upper surface of the table **904**. In other embodiments, the adjustable arm support **902** can be elevated from the stowed position to a position above an upper surface of the table **904**.

[0100] The adjustable arm support **902** can provide several degrees of freedom, including lift, lateral translation, tilt, etc. In the illustrated embodiment of FIGS. **9**A and **9**B, the arm support **902** is configured with four degrees of freedom, which are illustrated with arrows in FIG. **9**A. A first degree of freedom allows for adjustment of the adjustable arm support **902** in the z-direction (“Z-lift”). For example, the adjustable arm support **902** can include a carriage **906** configured to move up or down along or relative to a column **908** supporting the table **904**. A second degree of freedom can allow the adjustable arm support **902** to tilt. For example, the adjustable arm support **902** can include a rotary joint, which can allow the adjustable arm support **902** to be aligned with the bed in a Trendelenburg position.

[0101] A third degree of freedom can allow the adjustable arm support **902** to “pivot up,” which can be used to adjust a distance between a side of the table **904** and the adjustable arm support **902**. A fourth degree of freedom can permit translation of the adjustable arm support **902** along a longitudinal length of the table.

[0102] The surgical robotics system **900** in FIGS. **9**A and **9**B can comprise a table **904** supported by a column **908** that is mounted to a base **910**. The base **910** and the column **908** support the table **904** relative to a support surface. A floor axis **912** and a support axis **914** are shown in FIG. **9**B.

[0103] The adjustable arm support **902** can be mounted to the column **908**. In other embodiments, the arm support **902** can be mounted to the table **904** or the base **910**. The adjustable arm support **902** can include a carriage **906**, a bar or rail connector **916** and a bar or rail **918**. In some embodiments, one or more robotic arms mounted to the rail **918** can translate and move relative to one another.

[0104] The carriage **906** can be attached to the column **908** by a first joint **920**, which allows the carriage **906** to move relative to the column **908** (e.g., such as up and down a first or vertical axis **922**). The first joint **920** can provide the first degree of freedom (“Z-lift”) to the adjustable arm support **902**. The adjustable arm support **902** can include a second joint **924**, which provides the second degree of freedom (tilt) for the adjustable arm support **902**.

[0105] The adjustable arm support **902** can include a third joint **926**, which can provide the third degree of freedom (“pivot up”) for the adjustable arm support **902**. An additional joint **928** (shown in FIG. **9**B) can be provided that mechanically constrains the third joint **926** to maintain an orientation of the rail **918** as the rail connector **916** is rotated about a third axis **930**. The adjustable arm support **902** can include a fourth joint **932**, which can provide a fourth degree of freedom (translation) for the adjustable arm support **902** along a fourth axis **934**.

[0106] FIG. **9**C illustrates an end view of the surgical robotics system **900** with two adjustable arm supports **902***a* and **902***b* mounted on opposite sides of the table **904**. A first robotic arm **936***a* is attached to the first bar or rail **918***a* of the first adjustable arm support **902***a*. The first robotic arm **936***a* includes a base **938***a* attached to the first rail **918***a*. The distal end of the first robotic arm **936***a* includes an instrument drive mechanism or input **940***a* that can attach to one or more robotic medical instruments or tools. Similarly, the second robotic arm **936***b* includes a base **938***a* attached to the second rail **918***b*. The distal end of the second robotic arm **936***b* includes an instrument drive mechanism or input **940***b* configured to attach to one or more robotic medical instruments or tools.

[0107] In some embodiments, one or more of the robotic arms **936***a,b* comprises an arm with seven or more degrees of freedom. In some embodiments, one or more of the robotic arms **936***a,b* can include eight degrees of freedom, including an insertion axis (1-degree of freedom including insertion), a wrist (3-degrees of freedom including wrist pitch, yaw and roll), an elbow (1-degree of freedom including elbow pitch), a shoulder (2-degrees of freedom including shoulder pitch and yaw), and base **938***a,b* (1-degree of freedom including translation). In some embodiments, the insertion degree of freedom can be provided by the robotic arm **936***a,b*, while in other embodiments, the instrument itself provides insertion via an instrument-based insertion architecture.

C. Instrument Driver & Interface

[0108] The end effectors of a system's robotic arms comprise (i) an instrument driver (alternatively referred to as “tool driver,” “instrument drive mechanism,” “instrument device manipulator,” and “drive input”) that incorporate electro-mechanical means for actuating the medical instrument and, (ii) a removable or detachable medical instrument, which may be devoid of any electro-mechanical components, such as motors. This dichotomy may be driven by the need to sterilize medical instruments used in medical procedures, and the inability to adequately sterilize expensive capital equipment due to their intricate mechanical assemblies and sensitive electronics. Accordingly, the medical instruments may be designed to be detached, removed, and interchanged from the instrument driver (and thus the system) for individual sterilization or disposal by the physician or the physician's staff. In contrast, the instrument drivers need not be changed or sterilized, and may be draped for protection.

[0109] FIG. **10** illustrates an example instrument driver **1000**, according to one or more embodiments. Positioned at the distal end of a robotic arm, the instrument driver **1000** comprises of one or more drive output **1002** arranged with parallel axes to provide controlled torque to a medical instrument via corresponding drive shafts **1004**. Each drive output **1002** comprises an individual drive shaft **1004** for interacting with the instrument, a gear head **1006** for converting the motor shaft rotation to a desired torque, a motor **1008** for generating the drive torque, and an encoder **1010** to measure the speed of the motor shaft and provide feedback to control circuitry **1012**, which can also be used for receiving control signals and actuating the drive output **1002**. Each drive output **1002** being independent controlled and motorized, the instrument driver **1000** may provide multiple (at least two shown in FIG. **10**) independent drive outputs to the medical instrument. In operation, the control circuitry **1012** receives a control signal, transmits a motor signal to the motor **1008**, compares the resulting motor speed as measured by the encoder **1010** with the desired speed, and modulates the motor signal to generate the desired torque.

[0110] For procedures that require a sterile environment, the robotic system may incorporate a drive interface, such as a sterile adapter connected to a sterile drape, that sits between the instrument driver and the medical instrument. The chief purpose of the sterile adapter is to transfer angular motion from the drive shafts of the instrument driver to the drive inputs of the instrument while maintaining physical separation, and thus sterility, between the drive shafts and drive inputs. Accordingly, an example sterile adapter may comprise of a series of rotational inputs and outputs intended to be mated with the drive shafts of the instrument driver and drive inputs on the instrument. Connected to the sterile adapter, the sterile drape, comprised of a thin, flexible material such as transparent or translucent plastic, is designed to cover the capital equipment, such as the instrument driver, robotic arm, and cart (in a cart-based system) or table (in a table-based system). Use of the drape would allow the capital equipment to be positioned proximate to the patient while still being located in an area not requiring sterilization (i.e., non-sterile field). On the other side of the sterile drape, the medical instrument may interface with the patient in an area requiring sterilization (i.e., sterile field).

D. Medical Instrument

[0111] FIG. **11** illustrates an example medical instrument **1100** with a paired instrument driver **1102**. Like other instruments designed for use with a robotic system, the medical instrument **1100** (alternately referred to as a “surgical tool”) comprises an elongated shaft **1104** (or elongate body) and an instrument base **1106**. The instrument base **1106**, also referred to as an “instrument handle” due to its intended design for manual interaction by the physician, may generally comprise rotatable drive inputs **1108**, e.g., receptacles, pulleys or spools, that are designed to be mated with drive outputs **1110** that extend through a drive interface on the instrument driver **1102** at the distal end of a robotic arm **1112**. When physically connected, latched, and/or coupled, the mated drive inputs **1108** of the instrument base **1106** may share axes of rotation with the drive outputs **1110** in the instrument driver **1102** to allow the transfer of torque from the drive outputs **1110** to the drive inputs **1108**. In some embodiments, the drive outputs **1110** may comprise splines that are designed to mate with receptacles on the drive inputs **1108**.

[0112] The elongated shaft **1104** is designed to be delivered through either an anatomical opening or lumen, e.g., as in endoscopy, or a minimally invasive incision, e.g., as in laparoscopy. The elongated shaft **1104** may be either flexible (e.g., having properties similar to an endoscope) or rigid (e.g., having properties similar to a laparoscope) or contain a customized combination of both flexible and rigid portions. When designed for laparoscopy, the distal end of a rigid elongated shaft **1104** may be connected to an end effector of a surgical tool or medical instrument extending from a jointed wrist formed from a clevis with at least one degree of freedom, such as, for example, a grasper or scissors, that may be actuated based on force from the tendons as the drive inputs **1108** rotate in response to torque received from the drive outputs **1110** of the instrument driver **1102**. When designed for endoscopy, the distal end of the flexible elongated shaft **1104** may include a steerable or controllable bending section that may be articulated and bent based on torque received from the drive outputs **1110** of the instrument driver **1102**.

[0113] In some embodiments, torque from the instrument driver **1102** is transmitted down the elongated shaft **1104** using tendons along the shaft **1104**. These individual tendons, such as pull wires, may be individually anchored to individual drive inputs **1108** within the instrument handle **1106**. From the handle **1106**, the tendons are directed down one or more pull lumens along the elongated shaft **1104** and anchored at the distal portion of the elongated shaft **1104**, or in the wrist at the distal portion of the elongated shaft. During a surgical procedure, such as a laparoscopic, endoscopic, or a hybrid procedure, these tendons may be coupled to a distally mounted end effector, such as a wrist, a grasper, or scissors. Under such an arrangement, torque exerted on the drive inputs **1108** would transfer tension to the tendon, thereby causing the end effector to actuate in some way. In some embodiments, during a surgical procedure, the tendon may cause a joint to rotate about an axis, thereby causing the end effector to move in one direction or another. Alternatively, the tendon may be connected to one or more jaws of a grasper at distal end of the elongated shaft **1104**, where tension from the tendon cause the grasper to close.

[0114] In endoscopy, the tendons may be coupled to a bending or articulating section positioned along the elongated shaft **1104** (e.g., at the distal end) via adhesive, control ring, or other mechanical fixation. When fixedly attached to the distal end of a bending section, torque exerted on drive inputs **1108** would be transmitted down the tendons, causing the softer, bending section (sometimes referred to as the articulable section or region) to bend or articulate. Along the non-bending sections, it may be advantageous to spiral or helix the individual pull lumens that direct the individual tendons along (or inside) the walls of the endoscope shaft to balance the radial forces that result from tension in the pull wires. The angle of the spiraling and/or spacing there between may be altered or engineered for specific purposes, wherein tighter spiraling exhibits lesser shaft compression under load forces, while lower amounts of spiraling results in greater shaft compression under load forces, but also exhibits limits bending. On the other end of the spectrum, the pull lumens may be directed parallel to the longitudinal axis of the elongated shaft **1104** to allow for controlled articulation in the desired bending or articulable sections.

[0115] In endoscopy, the elongated shaft **1104** houses a number of components to assist with the robotic procedure. The shaft may comprise of a working channel for deploying surgical tools (or medical instruments), irrigation, and/or aspiration to the operative region at the distal end of the shaft **1104**. The shaft **1104** may also accommodate wires and/or optical fibers to transfer signals to/from an optical assembly at the distal tip, which may include of an optical camera. The shaft **1104** may also accommodate optical fibers to carry light from proximally-located light sources, such as light emitting diodes, to the distal end of the shaft.

[0116] At the distal end of the instrument **1100**, the distal tip may also comprise the opening of a working channel for delivering tools for diagnostic and/or therapy, irrigation, and aspiration to an operative site. The distal tip may also include a port for a camera, such as a fiberscope or a digital camera, to capture images of an internal anatomical space. Relatedly, the distal tip may also include ports for light sources for illuminating the anatomical space when using the camera.

[0117] In the example of FIG. **11**, the drive shaft axes, and thus the drive input axes, are orthogonal to the axis of the elongated shaft. This arrangement, however, complicates roll capabilities for the elongated shaft **1104**. Rolling the elongated shaft **1104** along its axis while keeping the drive inputs **1108** static results in undesirable tangling of the tendons as they extend off the drive inputs **1108** and enter pull lumens within the elongated shaft **1104**. The resulting entanglement of such tendons may disrupt any control algorithms intended to predict movement of the flexible elongated shaft during an endoscopic procedure.

[0118] FIG. **12** illustrates an alternative design for a circular instrument driver **1200** and corresponding instrument **1202** (alternately referred to as a “surgical tool”) where the axes of the drive units are parallel to the axis of the elongated shaft **1206** of the instrument **1202**. As shown, the instrument driver **1200** comprises four drive units with corresponding drive outputs **1208** aligned in parallel at the end of a robotic arm **1210**. The drive units and their respective drive outputs **1208** are housed in a rotational assembly **1212** of the instrument driver **1200** that is driven by one of the drive units within the assembly **1212**. In response to torque provided by the rotational drive unit, the rotational assembly **1212** rotates along a circular bearing that connects the rotational assembly **1212** to a non-rotational portion **1214** of the instrument driver **1200**. Power and control signals may be communicated from the non-rotational portion **1214** of the instrument driver **1200** to the rotational assembly **1212** through electrical contacts maintained through rotation by a brushed slip ring connection (not shown). In other embodiments, the rotational assembly **1212** may be responsive to a separate drive unit that is integrated into the non-rotatable portion **1214**, and thus not in parallel with the other drive units. The rotational assembly **1212** allows the instrument driver **1200** to rotate the drive units and their respective drive outputs **1208** as a single unit around an instrument driver axis **1216**.

[0119] Like earlier disclosed embodiments, the instrument **1202** may include an elongated shaft **1206** and an instrument base **1218** (shown in phantom) including a plurality of drive inputs **1220** (such as receptacles, pulleys, and spools) that are configured to mate with the drive outputs **1208** of the instrument driver **1200**. Unlike prior disclosed embodiments, the instrument shaft **1206** extends from the center of the instrument base **1218** with an axis substantially parallel to the axes of the drive inputs **1220**, rather than orthogonal as in the design of FIG. **11**.

[0120] When coupled to the rotational assembly **1212** of the instrument driver **1200**, the medical instrument **1202**, comprising instrument base **1218** and instrument shaft **1206**, rotates in combination with the rotational assembly **1212** about the instrument driver axis **1216**. Since the instrument shaft **1206** is positioned at the center of the instrument base **1218**, the instrument shaft **1206** is coaxial with the instrument driver axis **1216** when attached. Thus, rotation of the rotational assembly **1212** causes the instrument shaft **1206** to rotate about its own longitudinal axis. Moreover, as the instrument base **1218** rotates with the instrument shaft **1206**, any tendons connected to the drive inputs **1220** in the instrument base **1218** are not tangled during rotation. Accordingly, the parallelism of the axes of the drive outputs **1208**, the drive inputs **1220**, and the instrument shaft **1206** allows for the shaft rotation without tangling any control tendons.

[0121] FIG. **13** illustrates a medical instrument **1300** having an instrument based insertion architecture in accordance with some embodiments. The instrument **1300** (alternately referred to as a “surgical tool”) can be coupled to any of the instrument drivers discussed herein above and, as illustrated, can include an elongated shaft **1302**, an end effector **1304** connected to the shaft **1302**, and a handle **1306** coupled to the shaft **1302**. The elongated shaft **1302** comprises a tubular member having a proximal portion **1308***a* and a distal portion **1308***b*. The elongated shaft **1302** comprises one or more channels or grooves **1310** along its outer surface and configured to receive one or more wires or cables **1312** therethrough. One or more cables **1312** thus run along an outer surface of the elongated shaft **1302**. In other embodiments, the cables **1312** can also run through the elongated shaft **1302**. Manipulation of the cables **1312** (e.g., via an instrument driver) results in actuation of the end effector **1304**.

[0122] The instrument handle **1306**, which may also be referred to as an instrument base, may generally comprise an attachment interface **1314** having one or more mechanical inputs **1316**, e.g., receptacles, pulleys or spools, that are designed to be reciprocally mated with one or more torque couplers on an attachment surface of an instrument driver.

[0123] In some embodiments, the instrument **1300** comprises a series of pulleys or cables that enable the elongated shaft **1302** to translate relative to the handle **1306**. In other words, the instrument **1300** itself comprises an instrument-based insertion architecture that accommodates insertion of the instrument, thereby minimizing the reliance on a robot arm to provide insertion of the instrument **1300**. In other embodiments, a robotic arm can be largely responsible for instrument insertion.

E. Controller

[0124] Any of the robotic systems described herein can include an input device or controller for manipulating an instrument attached to a robotic arm. In some embodiments, the controller can be coupled (e.g., communicatively, electronically, electrically, wirelessly and/or mechanically) with an instrument such that manipulation of the controller causes a corresponding manipulation of the instrument e.g., via master slave control.

[0125] FIG. **14** is a perspective view of an embodiment of a controller **1400**. In the present embodiment, the controller **1400** comprises a hybrid controller that can have both impedance and admittance control. In other embodiments, the controller **1400** can utilize just impedance or passive control. In other embodiments, the controller **1400** can utilize just admittance control. By being a hybrid controller, the controller **1400** advantageously can have a lower perceived inertia while in use.

[0126] In the illustrated embodiment, the controller **1400** is configured to allow manipulation of two medical instruments, and includes two handles **1402**. Each of the handles **1402** is connected to a gimbal **1404**, and each gimbal **1404** is connected to a positioning platform **1406**.

[0127] As shown in FIG. **14**, each positioning platform **1406** includes a selective compliance assembly robot arm (SCARA) **1408** coupled to a column **1410** by a prismatic joint **1412**. The prismatic joints **1412** are configured to translate along the column **1410** (e.g., along rails **1414**) to allow each of the handles **1402** to be translated in the z-direction, providing a first degree of freedom. The SCARA arm **1408** is configured to allow motion of the handle **1402** in an x-y plane, providing two additional degrees of freedom.

[0128] In some embodiments, one or more load cells are positioned in the controller **1400**. For example, in some embodiments, a load cell (not shown) is positioned in the body of each of the gimbals **1404**. By providing a load cell, portions of the controller **1400** are capable of operating under admittance control, thereby advantageously reducing the perceived inertia of the controller **1400** while in use. In some embodiments, the positioning platform **1406** is configured for admittance control, while the gimbal **1404** is configured for impedance control. In other embodiments, the gimbal **1404** is configured for admittance control, while the positioning platform **1406** is configured for impedance control. Accordingly, for some embodiments, the translational or positional degrees of freedom of the positioning platform **1406** can rely on admittance control, while the rotational degrees of freedom of the gimbal **1404** rely on impedance control.

F. Navigation and Control

[0129] Traditional endoscopy may involve the use of fluoroscopy (e.g., as may be delivered through a C-arm) and other forms of radiation-based imaging modalities to provide endoluminal guidance to an operator physician. In contrast, the robotic systems contemplated by this disclosure can provide for non-radiation-based navigational and localization means to reduce physician exposure to radiation and reduce the amount of equipment within the operating room. As used herein, the term “localization” may refer to determining and/or monitoring the position of objects in a reference coordinate system. Technologies such as pre-operative mapping, computer vision, real-time EM tracking, and robot command data may be used individually or in combination to achieve a radiation-free operating environment. In other cases, where radiation-based imaging modalities are still used, the pre-operative mapping, computer vision, real-time EM tracking, and robot command data may be used individually or in combination to improve upon the information obtained solely through radiation-based imaging modalities.

[0130] FIG. **15** is a block diagram illustrating a localization system **1500** that estimates a location of one or more elements of the robotic system, such as the location of the instrument, in accordance to an example embodiment. The localization system **1500** may be a set of one or more computer devices configured to execute one or more instructions. The computer devices may be embodied by a processor (or processors) and computer-readable memory in one or more components discussed above. By way of example and not limitation, the computer devices may be in the tower **112** shown in FIG. **1**, the cart **102** shown in FIGS. **1-3**B, the beds shown in FIGS. **4-9**, etc.

[0131] As shown in FIG. **15**, the localization system **1500** may include a localization module **1502** that processes input data **1504***a*, **1504***b*, **1504***c*, and **1504***d* to generate location data **1506** for the distal tip of a medical instrument. The location data **1506** may be data or logic that represents a location and/or orientation of the distal end of the instrument relative to a frame of reference. The frame of reference can be a frame of reference relative to the anatomy of the patient or to a known object, such as an EM field generator (see discussion below for the EM field generator).

[0132] The various input data **1504***a-d* are now described in greater detail. Pre-operative mapping may be accomplished through the use of the collection of low dose CT scans. Pre-operative CT scans are reconstructed into three-dimensional images, which are visualized, e.g. as “slices” of a cutaway view of the patient's internal anatomy. When analyzed in the aggregate, image-based models for anatomical cavities, spaces and structures of the patient's anatomy, such as a patient lung network, may be generated. Techniques such as center-line geometry may be determined and approximated from the CT images to develop a three-dimensional volume of the patient's anatomy, referred to as model data **1504***a* (also referred to as “preoperative model data” when generated using only preoperative CT scans). The use of center-line geometry is discussed in U.S. patent application Ser. No. 14/523,760, the contents of which are herein incorporated in its entirety. Network topological models may also be derived from the CT-images, and are particularly appropriate for bronchoscopy.

[0133] In some embodiments, the instrument may be equipped with a camera to provide vision data **1504***b*. The localization module **1502** may process the vision data **1504***b* to enable one or more vision-based location tracking. For example, the preoperative model data may be used in conjunction with the vision data **1504***b* to enable computer vision-based tracking of the medical instrument (e.g., an endoscope or an instrument advance through a working channel of the endoscope). For example, using the preoperative model data **1504***a*, the robotic system may generate a library of expected endoscopic images from the model based on the expected path of travel of the endoscope, each image linked to a location within the model. Intra-operatively, this library may be referenced by the robotic system in order to compare real-time images captured at the camera (e.g., a camera at a distal end of the endoscope) to those in the image library to assist localization.

[0134] Other computer vision-based tracking techniques use feature tracking to determine motion of the camera, and thus the endoscope. Some features of the localization module **1502** may identify circular geometries in the preoperative model data **1504***a* that correspond to anatomical lumens and track the change of those geometries to determine which anatomical lumen was selected, as well as the relative rotational and/or translational motion of the camera. Use of a topological map may further enhance vision-based algorithms or techniques.

[0135] Optical flow, another computer vision-based technique, may analyze the displacement and translation of image pixels in a video sequence in the vision data **1504***b* to infer camera movement. Examples of optical flow techniques may include motion detection, object segmentation calculations, luminance, motion compensated encoding, stereo disparity measurement, etc. Through the comparison of multiple frames over multiple iterations, movement and location of the camera (and thus the endoscope) may be determined.

[0136] The localization module **1502** may use real-time EM tracking to generate a real-time location of the endoscope in a global coordinate system that may be registered to the patient's anatomy, represented by the preoperative model. In EM tracking, an EM sensor (or tracker) comprising of one or more sensor coils embedded in one or more locations and orientations in a medical instrument (e.g., an endoscopic tool) measures the variation in the EM field created by one or more static EM field generators positioned at a known location.

[0137] The location information detected by the EM sensors is stored as EM data **1504***c*. The EM field generator (or transmitter), may be placed close to the patient to create a low intensity magnetic field that the embedded sensor may detect. The magnetic field induces small currents in the sensor coils of the EM sensor, which may be analyzed to determine the distance and angle between the EM sensor and the EM field generator. These distances and orientations may be intra-operatively "registered" to the patient anatomy (e.g., the preoperative model) in order to determine the geometric transformation that aligns a single location in the coordinate system with a position in the pre-operative model of the patient's anatomy. Once registered, an embedded EM tracker in one or more positions of the medical instrument (e.g., the distal tip of an endoscope) may provide real-time indications of the progression of the medical instrument through the patient's anatomy.

[0138] Robotic command and kinematics data **1504***d* may also be used by the localization module **1502** to provide localization data **1506** for the robotic system. Device pitch and yaw resulting from articulation commands may be determined during pre-operative calibration. Intra-operatively, these calibration measurements may be used in combination with known insertion depth information to estimate the position of the instrument. Alternatively, these calculations may be analyzed in combination with EM, vision, and/or topological modeling to estimate the position of the medical instrument within the network.

[0139] As FIG. **15** shows, a number of other input data can be used by the localization module **1502**. For example, although not shown in FIG. **15**, an instrument utilizing shape-sensing fiber can provide shape data that the localization module **1502** can use to determine the location and shape of the instrument.

[0140] The localization module **1502** may use the input data **1504***a-d* in combination(s). In some cases, such a combination may use a probabilistic approach where the localization module **1502** assigns a confidence weight to the location determined from each of the input data **1504***a-d*. Thus, where the EM data **1504***c* may not be reliable (as may be the case where there is EM interference) the confidence of the location determined by the EM data **1504***c* can be decrease and the localization module **1502** may rely more heavily on the vision data **1504***b* and/or the robotic command and kinematics data **1504***d*.

[0141] As discussed above, the robotic systems discussed herein may be designed to incorporate a combination of one or more of the technologies above. The robotic system's computer-based control system, based in the tower, bed and/or cart, may store computer program instructions, for example, within a non-transitory computer-readable storage medium such as a persistent magnetic storage drive, solid state drive, or the like, that, upon execution, cause the system to receive and analyze sensor data and user commands, generate control signals throughout the system, and display the navigational and localization data, such as the position of the instrument within the global coordinate system, anatomical map, etc.

2. Introduction

[0142] Embodiments of the disclosure relate to systems and techniques for removably mounting an instrumentation core assembly of a surgical tool to a stage assembly of the surgical tool. The surgical tool may include a stage assembly and a core assembly that is removably mountable to the stage assembly. The stage assembly may include a lead screw and at least one spline extendable between first and second ends of the stage assembly, and a nut rotatably mounted to the lead screw to translate between the first and second ends upon rotation of the lead screw. The core assembly may include a drive housing, wherein the drive housing is movably mountable to the stage assembly to translate between the first and second end with the nut. The stage assembly may further include a carriage within which the nut is provided, such that the carriage translates axially between the first and second ends with the nut upon rotation of the lead screw, wherein the drive housing is removably coupled to the carriage to translate with the carriage. The core assembly may include an elongate shaft extending distally from the drive housing and penetrating the first end of the stage assembly when the drive housing is coupled to the stage assembly and an end effector arranged at a distal end of the elongate shaft. The surgical tool may include a primary shroud and a secondary shroud. The primary shroud may at least partially enclose the core assembly when installed on the stage assembly. The secondary shroud may be pivotably attached to the primary shroud at a hinge and movable between a closed position, where the secondary shroud occludes an opening in the primary shroud, and an open position, where the opening in the primary shroud is exposed. The secondary shroud may be slidingly attached to the primary shroud and circumferentially revolvable relative to the primary shroud between a closed position, where the secondary shroud occludes an opening in the primary shroud, and an open position, where the opening in the primary shroud is exposed. The secondary shroud may be slidingly attached to the primary shroud and axially slidable relative to the primary shroud between a closed position, where the secondary shroud occludes an opening in the primary shroud, and an open position, where the opening in the primary shroud is exposed.

3. Description

[0143] FIG. **16** is an isometric side view of an example surgical tool **1600** that may incorporate some or all of the principles of the present disclosure. The surgical tool **1600** may be similar in some respects to any of the medical instruments described above with reference to FIGS. **11-13** and, therefore, may be used in conjunction with a robotic surgical system, such as the robotically-enabled systems **100**, **400**, and **900** of FIGS. **1-13**. As illustrated, the surgical tool **1600** includes an elongated shaft **1602**, an end effector **1604** arranged at the distal end of the shaft **1602**, and an articulable wrist **1606** (alternately referred to as a “wrist joint”) that couples the end effector **1604** to the distal end of the shaft **1602**.

[0144] The terms “proximal” and “distal” are defined herein relative to a robotic surgical system having an interface configured to mechanically and electrically couple the surgical tool **1600** to a robotic manipulator. The term “proximal” refers to the position of an element closer to the robotic manipulator and the term “distal” refers to the position of an element closer to the end effector **1604** and thus closer to the patient during operation. Moreover, the use of directional terms such as above, below, upper, lower, upward, downward, left, right, and the like are used in relation to the illustrative embodiments as they are depicted in the figures, the upward or upper direction being toward the top of the corresponding figure and the downward or lower direction being toward the bottom of the corresponding figure.

[0145] The surgical tool **1600** can have any of a variety of configurations capable of performing one or more surgical functions. In the illustrated embodiment, the end effector **1604** comprises a surgical stapler, alternately referred to as an “endocutter,” configured to cut and staple (fasten) tissue. As illustrated, the end effector **1604** includes opposing jaws **1610**, **1612** configured to move (articulate) between open and closed positions. Alternatively, the end effector **1604** may comprise other types of instruments having the opposing jaws **1610**, **1612** such as, but not limited to, tissue graspers, surgical scissors, advanced energy vessel sealers, clip appliers, needle drivers, a babcock including a pair of opposed grasping jaws, bipolar jaws (e.g., bipolar Maryland grasper, forceps, a fenestrated grasper, etc.), etc. In other embodiments, the end effector **1604** may instead comprise any end effector or instrument capable of being operated in conjunction with the presently disclosed robotic surgical systems and methods. Such end effectors or instruments include, but are not limited to, a suction irrigator, an endoscope (e.g., a camera), or any combination thereof.

[0146] One or both of the jaws **1610**, **1612** may be configured to pivot to actuate the end effector **1604** between open and closed positions. In the illustrated example, the second jaw **1612** is rotatable (pivotable) relative to the first jaw **1610** to move between an open, unclamped position and a closed, clamped position. In other embodiments, however, the first jaw **1610** may move (rotate) relative to the second jaw **1612**, without departing from the scope of the disclosure. In yet other embodiments, both jaws **1610**, **1612** may move to actuate the end effector **1604** between open and closed positions.

[0147] In the illustrated example, the first jaw **1610** may be characterized or otherwise referred to as a “cartridge” or “channel” jaw, and the second jaw **1612** may be characterized or otherwise referred to as an “anvil” jaw. The first jaw **1610** may include a frame that houses or supports a staple cartridge, and the second jaw **1612** is pivotally supported relative to the first jaw **1610** and defines a surface that operates as an anvil to deform staples ejected from the staple cartridge during operation.
[0148] The wrist **1606** enables the end effector **1604** to articulate (pivot) relative to the shaft **1602** and thereby position the end effector **1604** at various desired orientations and locations relative to a surgical site. In the illustrated embodiment, the wrist **1606** is designed to allow the end effector **1604** to pivot (swivel) left and right relative to a longitudinal axis A**1** of the shaft **1602**. In other embodiments, however, the wrist **1606** may be designed to provide multiple degrees of freedom, including one or more translational variables (i.e., surge, heave, and sway) and/or one or more rotational variables (i.e., Euler angles or roll, pitch, and yaw). The translational and rotational variables describe the position and orientation of a component of a surgical system (e.g., the end effector **1604**) with respect to a given reference Cartesian frame. “Surge” refers to forward and backward translational movement, “heave” refers to translational movement up and down, and “sway” refers to translational movement left and right. With regard to the rotational terms, “roll” refers to tilting side to side, “pitch” refers to tilting forward and backward, and “yaw” refers to turning left and right.
[0149] In the illustrated embodiment, the pivoting motion at the wrist **1606** is limited to movement in a single plane, e.g., only yaw movement relative to the longitudinal axis A.sub.1. The end effector **1604** is depicted in FIG. **16** in the unarticulated position where a longitudinal axis of the end effector **1604** is substantially aligned with the longitudinal axis A.sub.1 of the shaft **1602**, such that the end effector **1604** is at a substantially zero angle relative to the shaft **1602**. In the articulated position, the longitudinal axis of the end effector **1604** would be angularly offset from the longitudinal axis A.sub.1 such that the end effector **1604** would be oriented at a non-zero angle relative to the shaft **1602**.
[0150] Still referring to FIG. **16**, the surgical tool **1600** may include a drive housing or “handle” **1614** that operates as an actuation system designed to facilitate articulation of the wrist **1606** and actuation (operation) of the end effector **1604** (e.g., clamping, firing, rotation, articulation, energy delivery, etc.). As described in more detail below, the handle **1614** includes coupling features that releasably couple the surgical tool **1600** to an instrument driver of a robotic surgical system.
[0151] The handle **1614** includes a plurality of drive members (obscured in FIG. **16**) that extend to the wrist **1606** and the end effector **1604**. Selective actuation of some drive members causes the end effector **1604** to articulate (pivot) relative to the shaft **1602** at the wrist **1606**. Selective actuation of other drive members cause the end effector **1604** to actuate (operate). Actuating the end effector **1604** may include closing and/or opening the second jaw **1612** relative to the first jaw **1610** (or vice versa), thereby enabling the end effector **1604** to grasp (clamp) onto tissue. Once tissue is grasped or clamped between the opposing jaws **1610**, **1612**, actuating the end effector **1604** may further include “firing” the end effector **1604**, which may refer to causing a cutting element or knife (not visible) to advance distally within a slot (obscured from view) defined in the first jaw **1610**. As it moves distally, the cutting element transects any tissue grasped between the opposing jaws **1610**, **1612**. Moreover, as the cutting element advances distally, a plurality of staples contained within the staple cartridge (e.g., housed within the first jaw **1610**) are urged (cammed) into deforming contact with corresponding anvil surfaces (e.g., pockets) provided on the second jaw **1612**. The deployed staples may form multiple rows of staples that seal opposing sides of the transected tissue.
[0152] As illustrated, the handle **1614** has a first or “distal” end **1618***a* and a second or “proximal” end **1618***b* opposite the first end **1618***a*. In some embodiments, one or more struts **1620** (two shown) extend longitudinally between the first and second ends **1618***a*,*b* to help fix the distance

between the first and second ends **1618***a,b*, provide structural stability to the handle **1614**, and secure the first end **1618***a* to the second end **1618***b*. In other embodiments, however, the struts **1620** may be omitted, without departing from the scope of the disclosure.

[0153] A lead screw **1622** and one or more splines **1624** also extend longitudinally between the first and second ends **1618***a,b*. In the illustrated embodiment, the handle **1614** includes a first spline **1624***a*, a second spline **1624***b*, and a third spline **1624***c*. While three splines **1624***a-c* are depicted, more or less than three may be included in the handle **1614**, without departing from the scope of the disclosure. Unlike the struts **1620**, the lead screw **1622** and the splines **1624***a-c* are rotatably mounted to the first and second ends **1618***a,b*. As described in more detail below, selective rotation of the lead screw **1622** and the splines **1624***a-c* causes actuation of various components within the handle **1614**, which thereby causes various functions of the surgical tool **1600** to transpire, for example, such as translating the end effector **1604** along the longitudinal axis A.sub.1, causing the end effector **1604** to articulate (pivot) at the wrist **1606**, and causing the end effector **1604** to actuate (operate).

[0154] The handle **1614** further includes a carriage or kart **1626** movably mounted along the lead screw **1622** and the splines **1624***a-c* and housing various activating mechanisms configured to cause operation of specific functions of the end effector **1604**. The carriage **1626** may comprise two or more layers, shown in FIG. **16** as a first layer **1628***a*, a second layer **1628***b*, a third layer **1628***c*, a fourth layer **1628***d*, and a fifth layer **1628***e*. While five layers **1628***a-e* are depicted, more or less than five may be included in the carriage **1626**, without departing from the scope of the disclosure.

[0155] In addition, one or more of the layers **1628***a-e* of the carriage **1626** may be detachable relative to the remaining layers **1628***a-e*. In this manner, one or more of the layers **1628***a-e* of the carriage **1626** may be secured together as a block of layers that may be releasably secured to the remaining layer(s) to define the carriage **1628**. For example, some of the layers **1628***a-e* may be integrally secured together in series, to form a block of layers, and such block of layers may be selectively secured to the remaining one or more layers **1628***a-e* that are not integrally secured to the block of layers. In the illustrated embodiment, the layers **1628***b-e* are secured to each other in series using one or more mechanical fasteners **1630** (one visible) extending between the second layer **1628***b* and the fifth layer **1628***e* and through coaxially aligned holes in each layer **1628***b-e* to form a block of layers configured to be releasably secured to the first layer **1628***a*, as described below. In the illustrated example, one or more snaps **1632** (two visible) are utilized to align (or clock) and releasably attach the block of layers **1628***b-e* to the first layer **1628***a*, as further described below, but various other types of one or more releasable connector(s) may be utilized.

[0156] While four layers **1628***b-e* are depicted as being secured together via the mechanical fastener(s) **1630** to define the block of secured-together layers **1628***b-e* releasably attached to the first layer **1628***a* via the snaps **1632**, the block of secured-together layers may include more or less than four layers, without departing from the scope of the disclosure. Moreover, one or more additional layers may be mechanically fastened in series to the first layer **1628***a* (e.g., the second layer **1628***b*) to define a second block of secured-together layers that is releasably attached to the first block of secured-together layers (e.g., the three layers **1628***c-e*), without departing from the scope of the disclosure. As further described below, configuring the carriage **1626** with one or more layers that are releasably secured relative to the remaining layers will provide a degree of modularity to the surgical tool **1600** to thereby allow one or more components of the surgical tool **1600** to be interchangeable, reusable, and/or replaceable.

[0157] The shaft **1602** is coupled to and extends distally from the carriage **1626** through the first end **1618***a* of the handle **1614**. In the illustrated embodiment, for example, the shaft **1602** penetrates the first end **1618***a* at a central aperture defined through the first end **1618***a*. The carriage **1626** is movable between the first and second ends **1618***a,b* along the longitudinal axis A.sub.1 and is thereby able to advance or retract the end effector **1604** relative to the handle **1614**, as indicated by the arrows B. More specifically, in some embodiments, the carriage **1626** includes a carriage nut

1634 mounted to the lead screw **1622** and secured with respect to the first layer **1628***a*. In this manner, the first layer **1628***a* of the carriage **1626** may define an elevator upon which the other layers **1628***b-e* releasably attached thereon may be translated, as described below. The outer surface of the lead screw **1622** defines helical threading and the carriage nut **1634** defines corresponding internal helical threading (not shown) matable with the outer helical threading of the lead screw **1622**. As a result, rotation of the lead screw **1622** causes the carriage nut **1634** to advance or retract the carriage **1626** along the longitudinal axis A.sub.1 and correspondingly advance or retract the end effector **1604** relative to the handle **1614**.

[0158] As indicated above, the lead screw **1622** and the splines **1624***a-c* are rotatably mounted to the first and second ends **1618***a,b*. More specifically, the first end **1618***a* of the handle **1614** may include one or more rotatable drive inputs, shown as a first drive input **1636***a*, a second drive input **1636***b*, a third drive input **1636***c*, and a fourth drive input **1636***d*. As discussed in more detail below, each drive input **1636***a-d* may be matable with a corresponding drive output of an instrument driver such that movement (rotation) of a given drive output correspondingly moves (rotates) the associated drive input **1636***a-d*.

[0159] The first drive input **1636***a* may be operatively coupled to the lead screw **1622** such that rotation of the first drive input **1636***a* correspondingly rotates the lead screw **1622**, which causes the carriage nut **1634** and the first layer **1628***a* constraining the carriage nut **1634** to advance or retract along the longitudinal axis A.sub.1, depending on the rotational direction of the lead screw **1622**. Moreover, as described herein the first layer **1628***a* of the carriage **1626** may be configured as an elevator (of the carriage **1626**) that translates the remaining layers **1628***b-e* of the carriage **1626** releasably connected to the first layer **1628***a*. Thus, when the remaining layers **1628***b-e* are installed on the first layer **1628***a*, to thereby define the carriage **1626** as depicted in FIG. **16**, rotation of the first drive input **1636***a* correspondingly rotates the lead screw **1622**, which causes the entirety of the carriage **1626** now fully coupled to the carriage nut **1634**, to advance or retract along the longitudinal axis A.sub.1, depending on the rotational direction of the lead screw **1622**.

[0160] The second drive input **1636***b* may be operatively coupled to the first spline **1624***a* such that rotation of the second drive input **1636***b* correspondingly rotates the first spline **1624***a*. In some embodiments, the first spline **1624***a* may be operatively coupled to a first activating mechanism **1638***a* of the carriage **1626**, and the first activating mechanism **1638***a* may be operable to open and close the jaws **1610**, **1612**. Accordingly, rotating the second drive input **1636***b* will correspondingly actuate the first activating mechanism **1638***a* and open or close the jaws **1610**, **1612**, depending on the rotational direction of the first spline **1624***a*. In addition, the second layer **1628***b* is configured to accommodate the first activating mechanism **1638***a* such that the jaws **1610**, **1612** are operable as described herein and, in the illustrated example, the first activating mechanism **1638***a* is at least partially constrained by the second layer **1628***b* and the third layer **1628***c*.

[0161] The third drive input **1636***c* may be operatively coupled to the second spline **1624***b* such that rotation of the third drive input **1636***c* correspondingly rotates the second spline **1624***b*. In some embodiments, the second spline **1624***b* may be operatively coupled to a second activating mechanism **1638***b* of the carriage **1626**, and the second activating mechanism **1638***b* may be operable to articulate the end effector **1604** at the wrist **1606**. Accordingly, rotating the third drive input **1636***c* will correspondingly actuate the second activating mechanism **1638***b* and cause the wrist **1606** to articulate in at least one degree of freedom, depending on the rotational direction of the second spline **1624***b*. In addition, the third layer **1628***c* is configured to accommodate the second activating mechanism **1638***b* for articulation of the wrist **1606** as described herein and, in the illustrated example, the second activating mechanism **1638***b* is at least partially constrained by the third layer **1628***c* and the fourth layer **1628***d*.

[0162] The fourth drive input **1636***d* may be operatively coupled to the third spline **1624***c* such that rotation of the fourth drive input **1636***d* correspondingly rotates the third spline **1624***c*. In some embodiments, the third spline **1624***c* may be operatively coupled to a third activating mechanism

1638*c* of the carriage **1626**, and the third activating mechanism **1638***c* may be operable to fire the cutting element (knife) at the end effector **1604**. Accordingly, rotating the fourth drive input **1636***d* will correspondingly actuate the third activating mechanism **1638***c* and cause the knife to advance or retract, depending on the rotational direction of the third spline **1624***c*. In addition, the fifth layer **1628***e* is configured to accommodate the third activating mechanism **1638***c* for firing the cutting element of the end effector **1604** as described herein and, in the illustrated example, the third activating mechanism **1638***c* is at least partially constrained by the fifth layer **1628***e* and a thrust bearing layer **1628***f* of the carriage **1626**.
[0163] In the illustrated embodiment, the activating mechanisms **1638***a-c* comprise intermeshed gearing assemblies including one or more drive gears driven by rotation of the corresponding spline **1624***a-c* and configured to drive one or more corresponding driven gears that cause operation of specific functions of the end effector **1604**.
[0164] FIG. **16** illustrates an embodiment of the handle **1614** having a shroud **1640** that defines a periphery of the handle **1614** for handling and manipulation by the operator or user. In the illustrated embodiment, the shroud **1640** is depicted as a transparent material (or at least a partially transparent material), such that the internal components of the handle **1614** are visible through the shroud **1640**. However, in other examples, the shroud **1640** need not be transparent. Where included, the shroud **1640** may be sized to receive the lead screw **1622**, the splines **1624***a-c*, and the carriage **1626**, as well as other internal components of the handle **1614**.
[0165] In some embodiments, the shroud **1640** may be incorporated in a shroud assembly of the handle **1614**. FIG. **17** is an isometric view of the surgical tool **1600** of FIG. **16** illustrating the handle **1614** when partially disassembled, according to one or more embodiments. In particular, FIG. **17** depicts an exemplary shroud assembly **1700** of the handle **1614** incorporating the shroud **1640** when disassembled from the remaining portion of the handle **1614**, according to one or more embodiments. In the illustrated embodiment, the shroud assembly **1700** defines a tubular or cylindrical structure having (i) a first end **1702***a* matable with the first end **1618***a* of the handle **1614** and (ii) a second end **1702***b* opposite the first end **1702***a* and matable with the second end **1618***b* of the handle **1614**. The carriage **1626**, the lead screw **1622**, and the splines **1624***a-c* may all be accommodated within the interior of the shroud **1640**, and the carriage **1626** may engage and ride on one or more rails **1704** (sometimes referred to as guide rails) coupled to the shroud **1640**. The rails **1704** extend longitudinally and parallel to the lead screw **1622**, and the rails **1704** are sized to be received within corresponding notches **1706** defined on the outer periphery of the carriage **1626** and, more particularly, on one or more of the layers **1628***a-e*. As the carriage **1626** translates along the longitudinal axis A.sub.1, the rails **1704** help maintain the angular position of the carriage **1626** and assume any torsional loading that would otherwise adversely affect the carriage **1626**. In addition, the shroud **1640** may include one or more alignment notches **1708** for aligning the second end **1618***b* on the shroud assembly **1700**. The rails **1704** may be fastened within an interior of the shroud **1640** and, in the illustrated examples, are coupled within the interior of the shroud **1640** via exterior rails **1710** positioned exterior the shroud **1640** and connected to the (guide) rails **1704** via a plurality of fasteners **1712** that extend through the shroud **1640**.
[0166] FIG. **18**A is an isometric view of the surgical tool **1600** of FIGS. **16** and **17** releasably coupled to an example instrument driver **1800**, according to one or more embodiments. The instrument driver **1800** may be similar in some respects to the instrument drivers **1102**, **1200** of FIGS. **11** and **12**, respectively, and therefore may be best understood with reference thereto. Similar to the instrument drivers **1102**, **1200**, for example, the instrument driver **1800** may be mounted to or otherwise positioned at the end of a robotic arm (not shown) and designed to provide the motive forces required to operate the surgical tool **1600**. Unlike the instrument drivers **1102**, **1200**, however, the shaft **1602** of the surgical tool **1600** extends through and penetrates the instrument driver **1800**.

[0167] The instrument driver **1800** has a body **1802** having a first or "proximal" end **1804***a* and a second or "distal" end **1804***b* opposite the first end **1804***a*. In the illustrated embodiment, the first end **1804***a* of the instrument driver **1800** is matable with the first end **1618***a* of the handle **1614**, and the shaft **1602** of the surgical tool **1602** extends into the first end **1804***a*, through the body **1802**, and distally from the second end **1804***b* of the body **1802**.

[0168] FIG. **18**B provides separated isometric end views of the first end **1804***a* of the instrument driver **1800** and the first end **1618***a* of the handle **1614** of the surgical tool **1600** of FIGS. **16** and **17**. With the jaws **1610**, **1612** closed, the shaft **1602** and the end effector **1604** are designed to penetrate the instrument driver **1800** by extending through a central aperture **1806** defined longitudinally through the body **1802** between the first and second ends **1804***a,b*. To angularly align the surgical tool **1600** with the instrument driver **1800** in a proper angular orientation, one or more alignment guides **1808** may be provided or otherwise defined within the central aperture **1806** and configured to engage one or more corresponding alignment features **1810** provided by the surgical tool **1600** (obscured from view, see FIG. **17**). In the illustrated embodiment, the alignment feature **1810** comprises a protrusion or projection defined on or otherwise provided by an alignment nozzle **1812** extending distally from the first end **1618***a* of the handle **1614**. In one or more embodiments, the alignment guide **1808** may comprise a curved or arcuate shoulder configured to receive the alignment feature **1810** as the shaft **1602** enters the central aperture **1806** and guide the surgical tool **1600** to a proper angular alignment with the instrument driver **1800** as the shaft **1602** is advanced distally through the central aperture **1806**.

[0169] In addition, one or more additional alignment features **1814** may be arranged within the central aperture **1806** and configured to mate with one or more corresponding recesses (not illustrated) provided on the alignment nozzle **1812** for ensuring the surgical tool **1600** is installed at a proper rotational alignment with respect to the instrument driver **1800**.

[0170] As illustrated, a drive interface **1816** is provided at the first end **1804***a* of the instrument driver **1800**, and a driven interface **1818** is provided at the first end **1618***a* of the handle **1614**. The driver and driven interfaces **1816**, **1818** may be configured to mechanically, magnetically, and/or electrically couple the handle **1614** to the instrument driver **1800**. To accomplish this, the driver and driven interfaces **1816**, **1818** may provide one or more matable locating features configured to secure the handle **1614** to the instrument driver **1800**. In the illustrated embodiment, for example, the drive interface **1816** provides one or more interlocking features **1820** (three shown) configured to locate and mate with one or more substantially complimentary shaped pockets **1822** (three shown) provided on the driven interface **1818**. The interlocking features **1820**, exemplified as bulbous protrusions, may be configured to align and mate with the pockets **1822** via an interference or snap fit engagement, for example.

[0171] The instrument driver **1800** also includes one or more drive outputs that extend through the drive interface **1816** to mate with the drive inputs **1636***a-d* provided on the driven face **1818** at the first end **1618***a* of the handle **1614**. More specifically, in the illustrated embodiment, the drive interface **1816** of the instrument driver **1800** includes a first drive output **1824***a* matable with the first drive input **1636***a*, a second drive output **1824***b* matable with the second drive input **1636***b*, a third drive output **1824***c* matable with the third drive input **1636***c*, and a fourth drive output **1824***d* matable with the fourth drive input **1636***d*. In some embodiments, as illustrated, the drive outputs **1824***a-d* may comprise splines designed to mate with corresponding splined receptacles on the drive inputs **1636***a-d*. Once properly mated, the drive inputs **1636***a-d* will share axes of rotation with the corresponding drive outputs **1824***a-d* to allow the transfer of rotational torque from the drive outputs **1824***a-d* to the corresponding drive inputs **1636***a-d*. In some embodiments, each drive output **1824***a-d* may be spring loaded and otherwise biased to spring outwards away from the drive interface **1816**. Each drive output **1824***a-d* may be capable of partially or fully retracting into the drive interface **1816**.

[0172] In some embodiments, the instrument driver **1800** may include additional drive outputs, depicted in FIG. **18**B as a fifth drive output **1824***e* and a sixth drive output **1824***f*. The fifth and sixth drive outputs **1824***e,f* may be configured and positioned to mate with additional drive inputs (not shown) of the handle **1614** to help undertake one or more additional functions of the surgical tool. In the illustrated embodiment, the handle **1614** does not include additional drive inputs, and the driven interface **1818** defines corresponding pockets **1826** configured to receive the fifth and sixth drive outputs **1824***e,f*.

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[0173] The surgical tool **1600** may be modular such that certain features of the surgical tool **1600** may be removed and replaced. In this manner, a portion of the surgical tool **1600** may be reusable for multiple surgical procedures, and thereby reduce capital expenditure.

[0174] For example, the shaft **1602**, the articulable wrist **1606**, and the end effector **1604** may be releasably coupled within the handle **1614** to be operable during a medical procedure and then removed from the handle **1614** following the procedure and replaced. In these embodiments, the layers of the carriage **1626** associated with various functions of the shaft **1602**, the articulable wrist **1606**, and the end effector **1604**, together with their associated activating mechanisms, may be assembled as one or more separate components that may be removably secured within the handle **1614**.

[0175] In some embodiments, the layers **1628***b-e* are secured to the shaft **1602** and the activating mechanisms **1638***a-c* are operatively housed within their associated layer **1628***b-e* so as to actuate the surgical tool **1600** as described herein. In the illustrated embodiment, the layers **1628***b-e* are fastened together with the mechanical fastener(s) **1630** to constrain the associated activating mechanisms **1638***a-c* in positions where they intermesh to provide functionality at the end effector **1604** and the wrist **1606**. Here, the shaft **1602**, the end effector **1604**, the articulable wrist **1606**, the layers **1628***b-e*, and the activating mechanisms **1638***a-c* are all secured together to define a disposable instrument portion of the surgical tool **1600**.

[0176] In these embodiments, the layers associated with performing other functionality of the surgical tool **1600** may be permanently secured within (or to) the handle **1614**, thereby defining a reusable portion of the surgical tool **1600** that can be employed in more than one surgical procedure, and into which the disposable instrument portion may be releasably installed. In the illustrated embodiment, the first layer **1628***a* and its associated drive mechanisms define this reusable portion and, because these components are configured to translate the shaft **1602** along the longitudinal axis A.sub.1, they also define a stage portion of the surgical tool **1600** into which the disposable instrument portion may be releasably installed.

[0177] Here, the first layer **1628***a* of the carriage **1626**, the lead screw **1622**, and the carriage nut **1634** are operatively secured within (or to) the handle **1614**, such that the first layer **1628***a* defines a type of “elevator” that is translationally driven and onto which the disposable instrument portion of the surgical tool **1600** may be releasably secured. Also, one or more of the splines **1624***a-c* may be operably secured within the handle **1614** to form part of the reusable stage portion. In some embodiments, the splines **1624***a-c* extend between the first and second ends **1618***a,b*, whereas in other embodiments, the splines **1624** may be configured as expandable or telescoping members extending between the first end **1618***a* and the first layer **1628***a*. In such embodiments, forces exhibited by the splines **1624***a-c* during rotation may be transferred through the first layer **1628***a* to the associated activating mechanisms **1638***a-c* in the disposable instrument portion to operate the surgical tool **1600**.

[0178] FIG. **19** illustrates the surgical tool **1600** configured with a stage portion **1902** and an instrument portion **1904** that may releasably attach to the stage portion **1902**, according to one or more embodiments. In this embodiment, the stage portion **1902** may be reused and the instrument portion **1904** may be removable and disposed of after a predetermined number of uses and then replaced with a new (or refurbished) instrument portion **1904**. In the illustrated embodiment, the instrument portion **1904** includes various component parts of a surgical stapler (e.g., an “endocutter”), but in other embodiments the stage portion **1902** may be configured to receive instrument portions relating to other types of surgical tools.

[0179] As illustrated, the stage portion **1902** may include a removable cap **1906** that may be removed from the shroud **1640** so that the instrument portion **1904** may be installed on or otherwise coupled to the stage portion **1902**. In some embodiments, the removable cap **1906** is removably attachable to the second end **1618***b* of the handle **1614** and removable to allow the instrument portion **1904** to be mated with a proximal side of the stage. Here, the instrument portion **1904** includes a handle assembly **1908** (alternatively referred to as a “handle drive assembly”) comprising the layers **1628***b-e*, the thrust bearing layer **1628***f*, and the associated activating mechanisms **1638***a-c* constrained thereby, and the shaft **1602** operably extends through a correspondingly sized central aperture in at least a portion of the handle assembly **1908**. In this embodiment, the handle assembly **1908** may be dropped onto a proximal side of the first layer **1628***a* (i.e., the elevator layer) after removal of the removable cap **1906**.

[0180] FIGS. **20**A-**20**B show detailed views of the handle assembly **1908** of the instrument portion **1904** and the first layer **1628***a* (i.e., the elevator) of the stage portion **1902**, according to one or more embodiments. Referring first to FIG. **20**A, illustrated is a bottom isometric view of the instrument handle assembly **1908** of the instrument portion **1904**. Here, the handle assembly **1908** includes an engagement end **2002** arranged to abut (or be in proximity to) the first layer **1628***a* (FIG. **20**B) when the carriage **1626** is assembled for use. As shown, the snaps **1632** extend distally from the engagement end **2002**, in a direction generally corresponding with the direction of the shaft **1602**. The snaps **1632** may include a bulbous portion **2004** extending from a leg portion **2006**, and may be biased radially inward towards the shaft **1602**, which may prove advantageous in allowing each bulbous portion **2004** to be retained within corresponding openings of the first layer **1628***a*. As briefly mentioned above, the snaps **1632** and their corresponding openings in the first layer **1628***a* may be arranged to clock or angularly align the handle assembly **1908** relative to the stage portion **1902** (FIG. **20**B), to thereby simplify proper installation.

[0181] FIG. **20**B is an enlarged isometric view of the proximal end of the first layer **1628***a*, according to one or more embodiments. In the illustrated embodiment, the first layer **1628***a* is configured as an elevator having a floor **2010** (i.e., the elevator floor) fastened to a base **2012**, and the carriage nut **1634** may be constrained between the floor **2010** and the base **2012**. In some embodiments, the floor **2010** and the base **2012** may comprise an integral, monolithic part. In such embodiments, the carriage nut **1634** may be formed between the floor **2010** and the base **2012**. Alternatively, the carriage nut **1634** may be positioned or defined within one or both of the coaxially aligned screw apertures **2020***d* defined in the floor **2010** and the base **2012** and through which the lead screw **1622** extends.

[0182] As illustrated, the floor **2010** includes one or more openings **2014** configured to align with corresponding bulbous portions **2004** (FIG. **20**A) of each of the snaps **1632** (FIG. **20**A). The openings **2014** correspond in size with the snaps **1632** such that they may receive and retain the bulbous portions **2004**. The engagement end **2002** (FIG. **20**A) may snap into the floor **2010** of the first layer **1628***a* (i.e., the elevator layer). In particular, the bulbous portions **2004** freely enter the corresponding openings **2014** in the floor **2010** when the floor **2010** slides proximally to a position where the openings **2014** are not constrained by the shroud **1640**, but the bulbous portions **2004** are pressed into and retained in the corresponding openings **2014** in the floor **2010** by the shroud **1640** as the floor **2010** slides distally into shroud **1640**. The shroud **1640** may help hold the snaps **1632** in an inward position where they are engaged by and retained within the corresponding openings **2014** in the floor **2010** of the first layer **1628***a*. Thus, the handle assembly **1908** cannot separate from the floor **2010** until the floor **2010** travels to a proximal position relative to the shroud **1640** sufficient to allow the snaps **1632** to deflect radially outward to a position where the bulbous portions **2004** are not retained within the corresponding openings **2014** of the floor **2010**. For example, the snaps **1632** of the handle assembly **1908** may be flared radially outward so as to slide into wider regions of the corresponding openings **2014** in the floor **2010** when the first layer **1628***a* has been translated distally to a position where the corresponding openings **2014** in the floor **2010** are not constrained by the shroud **1640**, and then the shroud **1640** may force the snaps **1632** into narrower regions of the corresponding openings **2014** as the handle assembly **1908** travels distally into the shroud **1640** to thereby lock the handle assembly **1908** to the first layer **1628***a*.

[0183] The splines **1624***a-c* may extend at least partially through the carriage **1626**. In the illustrated embodiment, the splines **1624***a-c* extend through spline apertures **2020***a-c* defined in the first layer **1628***a* and the lead screw **1622** extends through a screw aperture **2020***d* also defined in the first layer **1628***a* (see FIG. **20**B). In addition, the handle assembly **1908** includes spline passages **2022***a-c* (spline passage **2022***c* is occluded from view) and screw passage **2024** that extend through layers **1628***b-e*, and optionally through the thrust bearing layer **1628***f*, and align with the spline apertures **2020***a-c* and the screw apertures **2020***d* of the first layer **1628***a* such that the splines **1624***a-c* and the lead screw **1622** may extend through the handle assembly **1908** in addition to the first layer **1628***a*. Also, the drive mechanisms of the associated activating mechanisms **1638***a-c* may be keyed to the shape of the splines **1624***a-c* (FIG. **20**A) so as to transmit rotation from the splines **1624***a-c*. Here, gears **2026***a-c* (FIG. **20**A) of the associated activating mechanisms **1638***a-c* each have bores **2028** keyed to the cross section of the associated spline **1624***a-c* so as to transmit rotation from the spline **1624***a-c* to the associated gear **2026***a-c*.

[0184] In the illustrated example, the first layer **1628***a* (i.e., the elevator) may include a central opening **2016** arranged to receive the shaft **1602** and the end effector **1604** of the surgical tool **1600** as further described below.

[0185] FIG. **21** illustrates how the handle assembly **1908** of the instrument portion **1904** may be releasably attached to the stage portion **1902**, according to one or more embodiments. In the illustrated embodiment, the handle assembly **1908** is depicted in the process of being positioned on the first layer **1628***a* (i.e., the elevator floor) of the stage portion **1902**, with the shaft **1602** extending from the handle assembly **1908** and being inserted through the central opening **2016** of the first layer **1628***a* (i.e., the elevator). In some embodiments, the handle assembly **1908** may be configured to assist engaging the stage portion **1902**. For example, the lead screw **1622** may be threaded through a second carriage nut **2102** aligned with the screw passage **2024** and constrained within the handle assembly **1908**. Where included, the second carriage nut **2102** and or the screw passage **2024** may be utilized to help align (or clock) the handle assembly **1908** with respect to the first layer **1628***a* of the stage portion **1902**, such that the splines **1624***a-c* align with the corresponding activating mechanisms **1638***a-c* (FIG. **17**), and the second carriage nut **2102** may further add stability when the handle assembly **1908** is installed on the stage portion **1902**. In addition, during manufacture the (first) carriage nut **1634** and the second carriage nut **2102** can be aligned and engaged with the lead screw **1622** to help remove any drive slop that may otherwise exist in the surgical tool **1600** when assembled.

[0186] In some embodiments, an alignment table **2104** may be utilized to help align the instrument portion **1904** with the stage portion **1902**. FIG. **21** illustrates the alignment table **2104** (hereinafter, "the table **2104**") when removed from the handle assembly **1908** of the instrument portion **1904**. In the illustrated embodiment, the table **2104** includes four legs **2106***a-d* that align with the spline passages **2022***a-c* and the screw passage **2024**. Thus, the legs **2106***a-d* may extend through their respective spline passages **2022***a-c* and screw passage **2024**, which may help retain the associated activating mechanisms **1638***a-c*, and also help maintain alignment of the respective spline passages **2022***a-c* and screw passage **2024** for receiving the splines **1624***a-c* and the lead screw **1622**. After connecting the handle assembly **1908** of the instrument portion **1904** to the stage portion **1902**, the table **2104** may be removed, and the removable cap **1906** may then be secured on the shroud assembly **1700**. In some embodiments, the table **2104** includes just three of the legs **2106***a-c* which are associated with the splines **1624***a-c*. In these embodiments, the legs **2106***a-c* are equal in length, such that each leg **2106***a-c* engages its associated spline **1624***a-c* within the first layer

1628*a* (i.e., in the elevator floor) to maintain alignment therewith when the first layer **1628***a* retracts to position and align splines **1624***a-c* with the associated activating mechanisms **1636***a-c* (FIG. **17**).

[0187] FIG. **22** illustrates the instrument portion **1904** being assembled on the stage portion **1902**, according to one or more embodiments. As illustrated, the table **2104** is assembled on the table handle assembly **1908** of the instrument portion **1904** and the first layer **1628***a* has been moved proximally to a location where the floor **2010** extends outward of the shroud **1640** to expose the openings **2014** for receipt of the snaps **1632**. By moving the first layer **1628***a* proximally as shown, the openings **2014** are not constrained in part by the shroud **1640** and the bulbous portions **2004** of the snaps **1632** may deflect and enter the openings **2014**, thereby connecting the first layer **1628***a* of the carriage **1626** to the other layers **1628***b-e* of the carriage **1626**, such that the carriage **1626** is now assembled as an integral unit. When the carriage **1626** moves distally into the shroud **1640** to a position where the floor **2010** is located within the shroud **1640**, the openings **2014** will be constrained by the shroud **1640**, such that the shroud **1640** retains the bulbous portions **2004** of the snaps **1632** within the openings **2014** by inhibiting the snaps **1632** from deflecting radially outward and separating from the openings **2014**, as described above.

[0188] FIG. **22** also illustrates how the table **2104** may be utilized to align the handle assembly **1908** of the instrument portion **1904** with the stage portion **1902**, according to one or more embodiments. As illustrated, the legs **2106***a-d* of the table **2104** may be positioned to extend within the spline passages **2022***a-c* and the screw passage **2024**, and may be used to rotate the handle assembly **1908** into proper rotational alignment for receiving the splines **1624***a-c* and the lead screw **1622** of the stage portion **1902** of the surgical tool **1600**. As the splines **1624***a-c* and the lead screw **1622** enter their associated spline passages **2022***a-c* and screw passage **2024**, they may engage a distal end of the corresponding leg **2106***a-d* (of the table **2104**) to move (or push) the table **2104** from the handle assembly **1908** as the handle assembly **1908** is pushed proximally onto the stage portion **1902**. As a result, the table **2104** is gradually ejected and removed from the handle assembly **1908**.

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[0189] FIGS. **23**A-**23**H illustrate example operation and modularity of the surgical tool **1600** of FIG. **16**, according to one or more embodiments. In particular, FIGS. **23**A-**23**H depict an exemplary progression for installing (or interchanging) the instrument portion **1904** on the stage portion **1902** of the surgical tool **1600**.

[0190] Referring first to FIG. **23**A, depicted is a proximal end **2302** of the stage portion **1902** of the surgical tool **1600** prior to installation of the instrument portion **1904** of the surgical tool **1600**. The removable cap **1906** is depicted as having been installed on the proximal end **2302** of the stage portion **1902**. As mentioned herein, the first layer **1628***a* provides the elevator **2304** that is translationally driven and movable within the shroud **1640** along the longitudinal axis A.sub.1, and, when the handle assembly **1908** is installed on the elevator **2304**, the elevator **2304** carries the handle assembly **1908** as it translates axially. Here, the elevator **2304** is illustrated moving in a proximal direction towards the proximal end **2302**, as indicated by arrow P.

[0191] FIG. **23**B depicts the proximal end **2302** after the cap **1906** has been removed. By removing the cap **1906** from the shroud **1640**, a user may gain access to an interior of the stage portion **1902** in which the elevator **2304** resides and translates. After gaining access to the interior of the stage portion **1902**, the user may install the instrument portion **1904** of the surgical tool **1600** on the elevator **2304**, as described below. In at least some embodiments, the user may continue moving the elevator **2304** in a proximal direction, as indicated by the arrow P, to position where at least a portion of the elevator **2304** protrudes or extends out of the shroud **1640**.

[0192] FIG. **23**C illustrates the elevator **2304** having been proximally moved into a position where it partially protrudes from the shroud **1640**. In this illustration, the elevator **2304**, which includes the floor **2010** and the base **2012**, has been moved proximally to where the floor **2010** is positioned outside of the shroud **1640** while the base **2012** remains within the shroud **1640**. By moving the elevator **2304** in this manner, the openings **2014** in the floor **2010** are unconstrained by the shroud **1640** and open at a periphery of the floor **2010** such that the snaps **1632** (FIGS. **20**A and **23**E-**23**F) may be inserted therein, as generally described above. However, the elevator **2304** may be moved further proximally to where it further protrudes from the shroud **1640** or even fully protrudes from the shroud **1640**.

[0193] The splines **1624***a-c* may each extend the same distance. However, in some embodiments, at least some of the splines **1624***a-c* extend at least partially through the elevator **2304** for engagement with the activating mechanisms **1638***a-c* (FIG. **22**) of the handle assembly **1908** (FIG. **21**) when the handle assembly **1908** is mounted to the floor **2010** of the elevator **2304**. Here, the second and third splines **1624***b-c* protrude proximally from the floor **2010** when the elevator **2304** is positioned in this proximal position relative to the stage **1902** (FIGS. **23**C-**23**D), whereas the first spline **1624***a* extends substantially flush with a face of the floor **2010** but will protrude proximally from the face of the floor **2010** when the elevator **2304** is moved distally (e.g., FIG. **23**G). In other embodiments, however, each of the splines **1624***a-c* extends substantially flush with a face of the floor **2010** without protruding proximally from the face of the floor **2010** when the elevator **2304** is moved distally (e.g., FIG. **23**G).

[0194] FIG. **23**D illustrates the end effector **1604** of the instrument portion **1904** being moved distally towards the elevator **2304**, as indicated by an arrow D. The instrument portion **1904** of the surgical tool **1600** (FIG. **20**A) may be installed on the stage portion **1902** after moving the elevator **2304** into a position accessible to the handle assembly **1908** (FIG. **22**). In the illustrated embodiment, the elevator **2304** includes an opening **2306** sized to receive the end effector **1604** and the shaft **1602**. The user may continue moving the instrument portion **1904** distally D towards the elevator **2304** and insert the end effector **1604** and shaft **1602** of the instrument portion **1904** through the opening **2306** in the elevator **2304**.

[0195] FIG. **23**E illustrates the instrument portion **1904** being advanced onto the stage portion **1902**, according to one or more embodiments. In particular, FIG. **23**E illustrates the handle assembly **1908** after having inserted the end effector **1604** (FIG. **23**D) and the shaft **1602** through the opening **2306** in the elevator **2304** and after having moved the handle assembly **1908** into a position proximate to the elevator **2304**. In this embodiment, the table **2104** is assembled on the handle assembly **1908** to help align the activating mechanisms **1638***a-c* (FIG. **22**) of the instrument portion **1904** with the splines **1624***a-c* of the stage portion **1902**. The user may hold the table **2104** to advance and position the instrument portion **1904** relative to the stage portion **1902**. Regardless of whether the table **2104** is utilized, the end effector **1604** and the shaft **1602** are advanced through the elevator **2304**, and advanced distally through the stage portion **1902**, through the first end **1618***a*, and through the alignment nozzle **1812** (FIG. **18**B), as described above.

[0196] When the handle assembly **1908** is aligned with the elevator **2304**, the snaps **1632** will be in alignment with their corresponding openings **2014** peripherally arranged about the floor **2010** (FIGS. **21-22**), such that the instrument portion **1904** may be attached to the stage portion **1902** (FIG. **19**). As mentioned above, the table **2104** may be utilized to align the stage portion **1902** and the instrument portion **1904** of the surgical tool **1600** (FIGS. **21-22**). The table **2104** may be utilized to rotate the handle assembly **1908** about the longitudinal axis A.sub.1 (FIG. **21**) into a position where it is operatively aligned with the stage portion **1902**. For example, the user may rotate the handle assembly **1908** into alignment with the elevator **2304** by rotating the table **2104** such that the first leg **2106***a* (FIG. **21**) of the table **2104**, which is arranged in the first spline passage **2022***a* (FIG. **21**) defined in the handle assembly **1908**, aligns with the first spline **1624***a* (FIG. **21**); such that the second leg **2106***b* (FIG. **21**) of the table **2104**, which is arranged in the second spline passage **2022***b* (FIG. **21**) defined in the handle assembly **1908**, aligns with the second spline **1624***b* (FIG. **21**); such that the third leg **2106***c* (FIG. **21**) of the table **2104**, which is arranged in the third spline passage **2022***c* (FIG. **21**) defined in the handle assembly **1908**, aligns with the third spline **1624***c* (FIG. **21**); and such that the fourth leg **2106***d* (FIG. **21**) of the table **2104**, which is arranged in the screw passage **2024** (FIG. **21**) defined in the handle assembly **1908**, aligns with the lead screw **1622** (FIG. **21**).

[0197] The handle assembly **1908** may be further advanced onto the stage portion **1902** and releasably secured on the stage portion **1902** as depicted in FIG. **23**F. In particular, FIG. **23**F illustrates the handle assembly **1908** being further advanced onto the stage portion **1902** such that the snaps **1632** engage or are otherwise received within corresponding openings **2014** (FIG. **20**B) in the floor **2010** (FIGS. **20**A-**20**B). In this embodiment, the floor **2010** of the elevator **2304** is positioned outside of the shroud **1640** (FIG. **22**), such that the shroud **1640** does not constrain the openings **2014**. In this manner, the snaps **1632** may enter the corresponding openings **2014** in the floor **2010**. For example, the openings **2014** may include a chamfered lip configured to urge the bulbous portion **2004** of the snap **1632** outward as the snap **1632** is advanced into the opening **2014**, thereby causing the leg portion **2006** carrying the bulbous portion **2004** to deflect (or flex) radially outward (FIG. **20**A), and continual advancement of the snap **1632** towards its opening **2014** will radially expand the bulbous portion **2004** and the leg portion **2006** until the bulbous portion **2004** is in a position where the leg portion **2006** may bias (or snap) the bulbous portion **2004** within the opening **2014**. Engagement of the snaps **1632** within their corresponding openings **2014** results in the handle assembly **1908** being releasably secured on the stage portion **1902**. At this time, the user may manually remove the table **2104** (if utilized).

[0198] By coupling the handle assembly **1908** on the elevator **2304**, the elevator **2304** will carry the handle assembly **1908** as it translates along the longitudinal axis A.sub.1, such that the elevator **2304** and the handle assembly **1908** may translate together. As shown in FIG. **23**G, the elevator **2304** and the handle assembly **1908** carried thereby may be distally translated, as indicated by the arrow D, upon actuation of the lead screw **1622**. By retracting the elevator **2304** back within the shroud **1640**, the inner wall of the shroud **1640** locks or traps the bulbous portions **2004** of the snaps **1632** (FIG. **23**E) within their corresponding openings **2014** (FIG. **23**C), such that the snaps **1632** may only be removed from their openings **2014** when the elevator **2304** is at least partially advanced out of the shroud **1640** (FIG. **23**C).

[0199] The elevator **2304** and the handle assembly **1908**, when secured together in this manner, together define the carriage **1626**. As the carriage **1626** advances in the distal direction (arrow D), the splines **1624***a-c* and/or the lead screw **1622** may extend into and through the spline passages **2022***a-c* and/or the screw passage **2024** (FIGS. **21**-**22**), and continual distal advancement of the carriage **1626** may result in the table **2104** being ejected from the handle assembly **1908** as exemplified in FIG. **23**G. For example, the splines **1624***a-c* and the lead screw **1622** enter the spline passages **2022***a-c* and the screw passage **2024**, respectively, as the elevator **2304** pulls the handle assembly **1908** distally, the proximal ends of the splines **1624***a-c* and the lead screw **1622** (shown in FIGS. **23**C-**23**D) will contact the ends of the legs **2106***a-d* (FIG. **21**) of the table **2104** and thereby push the table **2104** out of the handle assembly **1908**. Thus, the table **2104** may be automatically removed as the elevator **2304** pulls the handle assembly **1908** distally. Finally, the cap **1906** may be placed back on the proximal end **2302** of the stage portion **1902**. FIG. **23**H depicts the cap **1906** re-installed on the proximal end **2302** of the stage portion **1902**.

[0200] FIG. **23**I is a schematic flowchart of an exemplary method **2310** for assembling and operating the surgical tool **1600** of FIG. **16**), according to one or more embodiments. In the illustrated embodiment, the method **2310** includes providing a surgical tool having a reusable stage portion, as at **2312**, and then installing the reusable portion of the surgical tool on a robotic instrument driver (e.g., the instrument driver **1102**, **1200**, **1800** of FIG. **11**, FIG. **12**, and FIGS. **18**A-**18**B, respectively), as at **2314**. As described herein, the reusable stage portion of the surgical tool may have an elevator floor on which a removable handle portion of surgical tool may be mounted, and the elevator floor may be actuated to carry the removable handle portion of the surgical tool in one or more directions. The removable handle portion may include an end effector that translates with the elevator floor. The removable handle portion of the surgical tool may be mounted on the reusable stage portion prior to installing the reusable stage portion on the robotic tool drive or may be mounted on the reusable stage portion after installing the reusable stage portion on the robotic tool drive.

[0201] In some embodiments, a method is provided for mounting the removable handle portion of the surgical tool on the reusable stage portion, which includes providing the reusable stage portion with the elevator floor in a proximal position suitable for receiving the removable handle portion, and may sometimes also include positioning the elevator floor in the proximal position suitable for receiving the removable handle portion and removing a top cap of the surgical tool to expose the elevator floor. Thus, the method **2310** may include the step of mounting the removable handle portion of the surgical tool on the reusable stage portion, as at **2316**, and such step may further include either the sub-step of providing the reusable stage portion with the elevator floor already in a suitable proximal position for receiving the removable handle portion or the sub-steps of positioning the elevator floor in the proximal position suitable for receiving the removable handle portion and removing a top cap of the surgical tool to expose the elevator floor.

[0202] When in the proximal position, the removable handle portion may be operatively mounted on the elevator floor such that drive input imparted in the reusable stage portion by the robotic tool drive is transferred to activating mechanisms in the removable portion configured to activate and/or move the end effector **1604**. Drive input forces may be imparted on the activating mechanisms of the removable portion via drive splines of the reusable stage portion and various configurations of the drive splines incorporated in the reusable spline portion, for example, telescoping drive splines that extend or retract with movement of the elevator floor and/or fixed length drive splines.

[0203] In embodiments where the surgical tool includes a shroud assembly, operatively mounting the removable handle portion on the elevator floor may include dropping the removable handle portion through the shroud assembly and onto the elevator floor. After operatively mounting the removable handle portion on the elevator floor, the removable cap may be placed on a proximal end. Accordingly, after the mounting step **2316**, the method **2310** may include the step of using the surgical tool. During a procedure, the removable handle portion may be removed from the reusable stage portion and optionally disposed of, and then a new removable handle portion (e.g., having a different end effector or function) may be operatively mounted on the elevator floor, for example, by dropping the new removable handle portion onto the elevator floor through the shroud assembly. In this manner, the reusable stage portion is usable in more than just one operation and/or on more than one patient, thereby justifying the use and expense of components and materials in manufacture that have superior durability, weight, and stiffness, etc. characteristics. Thus, the method **2310** may further include the steps of removing the removable handle portion of the surgical tool from the reusable stage portion, as at **2318**, and installing (mounting) a new removable handle portion of the surgical tool on the reusable stage portion, as at **2320**.

Removable Endcap

[0204] In the embodiments described above, the splines **1624***a-c* (FIG. **16**) and the lead screw **1622** (FIG. **16**) extend between the first and second ends **1618***a,b* (FIG. **16**). In these embodiments the splines **1624***a-c* and the lead screw **1622** extend from the first end **1618***a*, where they are operatively connected to the drive inputs **1636***a-d* of the surgical tool **1600** that are matable with and driven by a corresponding drive output **1824***a-f* of the instrument driver **1800**, such that movement (rotation) of a given drive output **1824***a-f* correspondingly moves (rotates) the associated drive input **1636***a-d*, which thereby moves (rotates) the splines **1624***a-c* and the lead screw **1622** associated therewith. Also in these embodiments, the splines **1624***a-c* and the lead screw **1622** extend and are operatively coupled to the cap **1906**.

[0205] FIGS. **24**A and **24**B illustrate front and rear perspectives, respectively, of the cap **1906** of FIG. **19**. In particular, FIG. **24**A illustrates an interior engagement side **2402** of the cap **1906**, according to one or more embodiments. The interior engagement side **2402** is configured to receive the ends of the splines **1624***a-c* and the lead screw **1622** when the cap **1906** is installed, and also configured to permit uncoupling from the splines **1624***a-c* and the lead screw **1622** for removal of the cap **1906**. Thus, the interior engagement side **2402** may include couplings (receptacles) for each of the splines and/or drive screws. In the illustrated embodiment, the interior engagement side **2402** includes three spline couplings **2404***a-c* arranged to correspond with the three splines **1624***a-c*. However, the interior engagement side **2402** may include more or less than the three spline couplings **2404***a-c*, for example, in embodiments having more or less than the three splines **1624***a-c*. Thus, additional locations **2404***d-f* may be configured to receive additional splines or drive elements. Here, hex head set screws **2405** are arranged at the additional locations **2404***d-f* as the illustrated embodiment utilizes the three spline couplings **2404***a-c*. In addition, the interior engagement side **2402** includes a stage coupling **2406** arranged to correspond with the drive screw **1622**.

[0206] The spline couplings **2404***a-c* and stage coupling **2406** are rotatable within the removable lid **1906**, such that they rotate with their corresponding spline **1624***a-c* and lead screw **1622** when engaged therewith by installing the removable cap **1906**. Also, any or all of the spline couplings **2404***a-c* and/or stage coupling **2406** may be keyed to the end geometry of their corresponding splines **1624***a-c* and/or lead screw **1622**. In some embodiments, the splines **1624***a-c* each include a square shaped end (not illustrated) and each spline coupling **2404***a-c* includes a recess correspondingly shaped to receive the particular end geometry of the corresponding spline **1624***a-c*, so as to ensure that the spline couplings **2404***a-c* rotate with their associated splines **1624***a-c* while minimizing relative slippage therebetween. It should be appreciated, however, that the splines **1624***a-c* and the associated spline couplings **2404***a-c* may be keyed with other geometries (e.g., triangular, polygonal, ovoid, etc.) and that each associated spline and coupling pair may be keyed with a geometry different from one or more of the other associated spline and coupling pair, without departing from the present disclosure. Also in the illustrated embodiment, the stage coupling **2406** is illustrated as a low friction thrust bearing keyed to an end geometry of the lead screw **1622**. In other embodiments, however, the lead screw **1622** may be differently connected to the screw coupling **2406**, for example, in at least some embodiments, the lead screw **1622** may be threaded into the screw coupling **2406**.

[0207] In the illustrated embodiment, the removable cap **1906** further includes a frame assembly **2408** and a ring **2410** arranged about the frame assembly **2408**. The frame assembly **2408** is configured to retain the spline couplings **2404***a-c* and the stage coupling **2406**. In some examples, additional spline couplings (e.g., similar to the spline couplings **2404***a-c*) may be arranged at the additional locations **2404***d-f* with an organization that comports with a standard alignment of splines such that the removable cap **1906** is utilizable with the maximum number of splines even where the handle assembly **1908** riding thereon may not be configured to receive input from one or more of the maximum number of splines.

[0208] The interior engagement side **2402** of the removable cap **1906** may be configured to mate with the shroud assembly **1700** (FIG. **17**). In the illustrated example, the frame assembly **2408** includes a boss **2412** protruding outward to define a plurality of alignment tabs **2414**. When installing the removable cap **1906**, the alignment tabs **2414** may be used to ensure proper alignment by locating the alignment tabs **2414** within the corresponding alignment notches **1708** defined in the shroud **1640**.

Telescoping Splines

[0209] FIGS. **25**A and **25**B are partial cross-sectional side views of one or more example embodiments of the splines **1624***a-c* of FIGS. **16** and **17**, according to embodiments of the present disclosure. In at least some embodiments, one or more of the splines **1624***a-c* (and/or the lead screw **1622**) are configured to extend less than the entire distance between the first and second ends **1618***a,b*. For example, rather than being constrained between the first end **1618***a* and the removable cap **1906** as described above, one or more of the splines **1624***a-c* may be constrained between the first end **1618***a* and the carriage **1626**. In these embodiments, one or more of the splines **1624***a-c* may be configured to telescope such that it/they expand or contract as the elevator **2304** of the carriage **1626** translates.

[0210] FIGS. **25**A and **25**B illustrate the splines **1624***a-c* as telescoping structures. In the illustrated embodiment, the splines **1624***a-c* are configured to nest within a handle interface **2502** of the surgical tool **1600** (FIG. **16**). As illustrated, the handle interface **2502** is a structure of the surgical tool **1600** within which the one or more rotatable drive inputs (e.g., drive inputs **1636***a-d*) are operatively arranged for receiving input (e.g., rotational input) from the rotatable drive outputs (e.g., drive outputs **1824***a-d*) of the instrument driver **1800** (FIG. **18**B). Also, the splines **1624***a-c* are operatively connected to the drive inputs **1636***a-d* within the handle interface **2502**. Here, the handle interface **2502** includes a floor **2504**, and the splines **1624***a-c* extend from their respective rotatable drive inputs **1636***a-d*, upward through the floor **2504**, and to the elevator **2304**.

[0211] In the illustrated embodiment, the splines **1624***a-c* each comprise a series of telescoping portions **2506** configured to nest within the handle interface **2502**. FIG. **25**A illustrates the carriage **1626** having been translated into a distal most position to thereby cause the telescoping portions **2506** of the splines **1624***a-c* to substantially retract and nest within the handle interface **2502**. In this manner, the splines **1624***a-c* may be in a retracted or nested condition where portions of the splines **1624***a-c* are nested beneath the floor **2504** when the elevator **2304** has translated into a distal most position.

[0212] FIG. **25**B illustrates the elevator **2304** having been at least partially translated in a proximal direction to thereby cause the series of telescoping portions **2506** to telescope (or elongate, expand, extend, un-nest, etc.) into an at least partially extended state. In this manner, one or more of the series of telescoping portions **2506** may expand (extend) through the floor **2504** and into a stage chamber **2508** within which the carriage **1626** translates as the elevator **2304** pulls (or carries, expands, telescopes, un-nests, etc.) the splines **1624***a-c* into their extended state.

[0213] The series of telescoping portions **2506** may each include individual telescoping spline portions **2510***a-d*. The telescoping spline portions **2510***a-d* are operatively connected in series to transmit torque imparted on the rotatable drive inputs (e.g., the drive inputs **1636***b-d*) by the instrument driver **1800** to a coupling **2512**. Thus, the telescoping spline portions **2510***a-d* (of each spline **1624***a-c*) may be rotationally fixed relative to each other about an associated spline axis S.sub.1, S.sub.2, S.sub.3. In addition, the telescoping spline portions **2510***a-d* of each spline **1624***a-c* are serially connected such that they may telescope within each other.

[0214] Here, the first telescoping spline portion **2510***a* exhibits the largest diameter, and the second telescoping spline portion **2510***b* exhibits the second largest diameter and is slidingly retained within the first telescoping spline portion **2510***a*. The third telescoping spline portion **2510***c* exhibits the third largest diameter and is slidingly retained within the second telescoping spline portion **2510***b*, and the fourth telescoping spline portion **2510***d* exhibits the fourth largest diameter and is slidingly retained within the third telescoping spline portion **2510***c*. In this manner, the fourth telescoping spline portion **2510***d* may slide into (or nest within) the third telescoping spline portion **2510***c*, the third telescoping spline portion **2510***c* may slide into (or nest within) the second telescoping spline portion **2510***b*, the second telescoping spline portion **2510***b* may slide into (or nest within) the first telescoping spline portion **2510***a*, and the telescoping spline portions **2510***a-d* may nest together as the elevator **2304** moves distally towards the floor **2504**. Here, the telescoping spline portions **2510***a-d* nest in the handle interface **2502** such that the splines **1624***a-c* are concealable within the handle interface **2502**, but in other embodiments, they may be differently configured, for example, to nest within the elevator **2304**.

[0215] In some embodiments, the coupling **2512** may include a receptacle **2514** configured to transmit torque to a corresponding activating mechanism **1638***a-c* configured to drive and cause operation of specific functions of the end effector **1604**. In some embodiments, removable splines **2516***a-c* may be provided in the handle assembly **1908** for receiving drive input from the couplings

2512 and transmitting that imparted torque to the associated activating mechanisms **1638***a-c*, such that torque imparted by the instrument driver **1800** (FIGS. **18**A-**18**B) is operatively transferred to the handle assembly **1908** to permit various movements and actions of the end effector **1604**. Here, the removable splines **2516***a-c* may each include a torsion shaft **2520** having an input end **2522** positioned to communicate or mate with the coupling **2512** via the receptacle **2514**, which may rotationally lock the torsion shaft **2520** of each removable spline **2516***a-c* with the associated coupling **2512** and series of telescoping portions **2506**. For example, the receptacle **2514** may be keyed to the particular spline geometry of the input end **2522** associated with it. Also, each torsion shaft **2520** may include an output **2524** in operable engagement with the associated activating mechanisms **1638***a-c*. In some examples, the output **2524** is an output or drive gear that is rotationally fixed on the torsion shaft **2520** such that they rotate in unison.

[0216] FIG. **25**B depicts the carriage **1626** partially disassembled (i.e., with the handle assembly **1908** partially removed from the elevator **2304**), so as to more fully illustrate orientation and interaction of components within the handle assembly **1908**. For example, FIG. **25**B illustrates the telescoping nature of the series of telescoping portions **2506**, the orientation of the removable splines **2516***a-c* and the associated activating mechanisms **1638***a-c*, engagement of the removable splines **2516***a-c* of the handle assembly **1908** with the couplings **2512** of the elevator **2304**, etc. In this example, the first layer **1628***a* is the elevator **2304** on which the structural layers **1628***b-d* are releasably attachable and is associated with controlling axial translation of the carriage of the surgical tool **1600**. The second layer **1628***b* is the bottom layer of the handle assembly **1908** and is associated with controlling a first functionality of the end effector **1604**, the third layer **1628***c* is the middle layer of the handle assembly **1908** and is associated with controlling a second functionality of the end effector **1604**, and the fourth layer **1628***d* is the upper layer of the handle assembly **1908** and is associated with controlling a third functionality of the end effector **1604**.

[0217] In the illustrated example, the shroud **1640** is illustrated as a rigid tube member that does not expand or contract. Thus, in some examples the stage chamber **2508** defined by the shroud **1640** may be of fixed volume. However, in other examples, the shroud **1640** may be configured to expand or contract with movement of the elevator **2304** and telescoping splines **1624***a-c*. For example, the shroud **1640** may comprise a series of telescoping shroud portions arranged to telescope between nested and un-nested positions with movement of the elevator **2304**.

[0218] FIGS. **26**A and **26**B illustrate the splines **1624***a-c* configured to telescope, according to one or more alternate embodiments of the present disclosure. In the illustrated embodiment, the splines **1624***a-c* are provided as a series of telescoping spline members **2602** arranged to telescope from a nested or retracted position (FIG. **26**A) into one or more extended positions (FIG. **26**B). Thus, in this example, the splines **1624***a-c* are concealable within the elevator **2304**.

[0219] Here, the series of telescoping spline members **2602** are configured to telescope into the elevator **2304**. For example, each of the series of telescoping portions **2602** may include individual telescoping spline portions **2604***a-d*. The telescoping spline portions **2604***a-d* are operatively connected in series to transmit torque imparted on the rotatable drive inputs (e.g., the drive inputs **1636***b-d*) by the instrument driver **1800** to a coupling **2606**. Thus, the telescoping spline portions **2604***a-d* (of each spline **1624***a-c*) may be rotationally fixed relative to each other about an associated spline axis S.sub.1, S.sub.2, and S.sub.3. In addition, the telescoping spline portions **2604***a-d* (of each spline **1624***a-c*) are serially connected such that they may telescope within each other.

[0220] Here, the first telescoping spline portion **2604***a* exhibits the largest diameter and is at least partially arranged in or otherwise coupled to the elevator **2304**, and the second telescoping spline portion **2604***b* exhibits the second largest diameter and is slidingly retained within the first telescoping spline portion **2604***a*. The third telescoping spline portion **2604***c* exhibits the third largest diameter and is slidingly retained within the second telescoping spline portion **2604***b*, and the fourth telescoping spline portion **2604***d* exhibits the fourth largest diameter and is slidingly retained within the third telescoping spline portion **2604***c*. In this manner, the fourth telescoping spline portion **2604***d* may slide into (or nest within) the third telescoping spline portion **2604***c*, the third telescoping spline portion **2604***c* may slide into (or nest within) the second telescoping spline portion **2604***b*, the second telescoping spline portion **2604***b* may slide into (or nest within) the first telescoping spline portion **2604***a*, and the telescoping spline portions **2604***a-d* may nest together as the elevator **2304** moves distally towards the floor **2504**. Thus, the telescoping spline portions **2604***a-d* are shown configured to nest in the first frame layer **1628***a* (i.e., the elevator layer **2304**), but in other embodiments they may be configured to nest differently.

[0221] In some embodiments, the lead screw **1622** is a rigid member having a non-telescoping configuration. In other embodiments, however, the lead screw **1622** may be configured to telescope with the elevator **2304**. For example, the lead screw **1622** may be a telescoping linear actuator having a centermost telescoping member terminating on the first layer **1628***a* (i.e., the elevator floor).

[0222] FIG. **27** is a top, cross-sectional view of a telescoping spline configuration **2700** according to one or more additional embodiments. The splines **1624***a-c* (FIGS. **26**A-**26**B) may have varying telescoping configurations, such as the telescoping spline configuration **2700** exemplified in FIG. **27**. In the illustrated embodiment, the telescoping spline configuration **2800** includes a series of concentric spline portions having corresponding keyed geometries, so as to rotate together and transmit a torsional force through the telescoping spline configuration **2800**.

[0223] FIGS. **28**A-**28**B illustrate an expandable and collapsible shroud **2802** in respective collapsed and expanded positions, according to one or more embodiments. As mentioned above, the shroud **1640** may be configured to expand or contract with the elevator **2304**. In the illustrated embodiment, the expandable and collapsible shroud **2802** includes an expandable portion **2804** arranged to expand or contract with the elevator **2304**. The expandable portion **2804** may have various configurations, such as a bellows or accordion-like configuration, a telescoping configuration, etc.

Drive Puck for Translating System with Mapped Instrument Drive Outputs

[0224] FIG. **29** illustrates an elevator of the surgical tool **1600** that operates as a translatable drive puck (or nut) **2900**, according to one or more embodiments of the present disclosure. In these embodiments, the drive puck **2900** not only functions as a stage for the surgical tool **1600**, but also maps or transfers (e.g., via gearing) input forces received from the splines **1624***a-c* to predetermined drive locations within the envelope of the shroud **1640** that do not correspond (or are unassociated) with locations of the associated splines **1624**. As will be appreciated, this can allow utilization of differently shaped shrouds or shroud assemblies.

[0225] As illustrated, the drive puck **2900** is configured to translate on the splines **1624***a-c* within the shroud **1640** and transfer drive input forces received therefrom to off-set or remote drive output locations. The drive puck **2900** rides or slides on the splines **1624***a-c* within the shroud **1640** and is thus a translatable drive puck that may function as the stage (elevator) of the surgical tool **1600**. In the illustrated example, the drive puck **2900** translates along the splines **1624***a-c* via actuation of the lead screw **1622**. Thus, the drive puck **2900** may include a drive screw portion (obscured from view) that intermeshes with the lead screw **1622**. In some embodiments, for example, the carriage nut **1634** (FIG. **16**) may be constrained within a body **2902** of the drive puck **2900**. However, in other examples, a threaded bore **2904** (occluded, see FIGS. **30**A-**30**B and FIG. **31**) may be formed in the body **2902** and configured to receive the lead screw **1622** and cause translation of the drive puck **2900** upon rotation of the lead screw **1622**.

[0226] While the drive puck **2900** is configured to translate axially within the shroud **1640**, rotation of the drive puck **2900** within the shroud **1640** is inhibited. In the illustrated example, the drive puck **2900** defines or otherwise provides one or more guide notches **2906** sized to receive the rails **1704** such that the rails **1704** may ride within the notches **2906** as the drive puck **2900** translates within the shroud **1640**. Consequently, the rails **1704** guide the drive puck **2900** as it translates axially within the shroud **1640** and also constrain the drive puck **2900** from rotating by countering any rotational forces acting on the drive puck **2900**. In this manner, any rotational force acting on the drive puck **2900** will be assumed by the rails **1704**, such that the rotational force is not imparted to the splines **1624***a-c* extending through the drive puck **2900**.

[0227] As illustrated, the splines **1624***a-c* extend through the drive puck **2900** and the drive puck **2900** is able to slide on or otherwise axially traverse the splines **1624***a-c*. Here, the drive puck **2900** is illustrated and described as being operable with three of the splines **1624***a-c*, however, the drive puck **2900** may be differently configured so that it is operable using input from any number and/or arrangement of splines. The drive puck **2900** may translate on the splines **1624***a-c* both when the splines **1624***a-c* are stationary and when one or more of the splines **1624***a-c* are activated (i.e., rotated via the instrument driver **1800**). To accomplish this, as illustrated, the drive puck **2900** may include spline couplings **2910***a-c* that correspond with the splines **1624***a-c* and are rotatable with the associated spline **1624***a-c*. Here, the first spline **1624***a* extends through the first spline coupling **2910***a*, the second spline **1624***b* extends through the second spline coupling **2910***b*, and the third spline **1624***c* extends through the third spline coupling **2910***c*. The spline couplings **2910***a-c* may each include an interior geometry configured to engage and mate with the geometry of the associated spline **1624***a-c*. In some embodiments, the spline couplings **2910***a-c* may each include a geometry keyed to the cross-section of the associated spline **1624***a-c*. In the illustrated embodiments, the spline couplings **2910***a-c* each define (exhibit) a square-shaped geometry matable with the associated spline **1624***a-c* such that rotation of the spline **1624***a-c* correspondingly rotates the associated spline coupling **2910***a-c*.

[0228] According to embodiments of the present disclosure, the drive puck **2900** may be configured such that the spline couplings **2910***a-c* transfer drive input forces to locations on the drive puck **2900** that are distant or positioned away (laterally offset) from the spline couplings **2910***a-c*. In particular, the spline couplings **2910***a-c* receive drive input forces from the associated splines **1624***a-c*, and internal gearing within the drive puck **2900** maps or transmits those drive input forces to corresponding drive outputs (e.g., drive pegs) at locations that do not correspond (align) with the spline couplings **2910***a-c* and the associated splines **1624***a-c*. Thus, the drive puck **2900** may receive drive input from the splines **1624***a-c* extending through an input region of the drive puck **2900** and transfer that drive input to any output region of the drive puck **2900** that is positioned away from the input region and configured to receive a removable handle drive assembly of the surgical tool **1600** as described herein.

[0229] In the illustrated example, the drive puck **2900** is cylindrically circular in shape and thus includes a generally circular proximal face (see FIG. **30**A) and a generally circular distal face (see FIG. **30**B). Here, the spline couplings **2910***a-c* are arranged on a first arcuate portion (e.g., a first half-arc-length) of the circular shaped distal and proximal faces of the drive puck **2900**. In addition, the drive puck **2900** includes one or more instrument drive outputs **2912***a-c* operatively connected with an associated one of the spline couplings **2910***a-c*. Here, the instrument drive outputs **2912***a-c* comprise drive pegs and are arranged on a second arcuate portion (e.g., a second half-arc-length) of the drive puck **2900**. In one example, the instrument drive outputs **2912***a-c* are provided as drive pegs with stub shafts extending (or protruding) outward from the proximal face of the drive puck **2900**.

[0230] In the illustrated example, the first spline coupling **2910***a* is operatively connected to the first instrument drive output **2912***a* such that rotation of the first spline coupling **2910***a* (via the first spline **1624***a*) rotates the first instrument drive output **2912***a*. The second spline coupling **2910***b* is operatively connected to the second instrument drive output **2912***b* such that rotation of the second spline coupling **2910***b* (via the second spline **1624***b*) rotates the second instrument drive output **2912***b*. The third spline coupling **2910***c* is operatively connected to the third instrument drive output **2912***c* such that rotation of the third spline coupling **2910***c* (via the third spline **1624***c*) rotates the third instrument drive output **2912***c*. As more fully described below, internal gearing may be provided within the drive puck **2900** to operatively intermesh each of the spline couplings **2910***a-c* with the instrument drive output **2912***a-c* associated therewith. In this manner, the drive puck **2900** receives drive input forces through the spline couplings **2910***a-c* interacting with the splines **1624***a-c* in the first hemisphere, and internal gearing within the drive puck **2900** transfers (or maps) each drive input force to one of the instrument drive outputs **2912***a-c* located in the second hemisphere. Thus, the instrument drive outputs **2912***a-c* may be arranged at locations about the drive puck **2900** that are independent of the orientation of the splines **1624***a-c* and/or at locations about the drive puck **2900** that do not correspond with the particular location(s) at which their associated splines **1624***a-c* engage the drive puck **2900**.

[0231] FIGS. **30**A and **30**B are isometric top and bottom views, respectively, of the drive puck **2900** of FIG. **29**, according to one or more embodiments. More specifically, FIG. **30**A depicts a first or "proximal" face **3002**, and FIG. **30**B depicts a second or "distal" face **3004** opposite the proximal face **3002**. As illustrated, the drive puck **2900** defines a central aperture or bore **3006** extending through the body **2902** between the proximal and distal faces **3002**, **3004**. The bore **3006** may be sized to receive the shaft **1602** of the surgical tool **1600** (FIG. **16**) when a handle drive assembly is installed thereon. In the illustrated embodiment, the body **2902** is hollow and defines an interior volume **3008** that is in communication with the bore **3006**.

[0232] In some embodiments, as illustrated, the spline couplings **2910***a-c* may include stub portions **3010** that extend or protrude from at least the proximal face **3002** of the body **2902**. In other examples, one or more of the stub portions **3010** may instead be arranged to extend from the distal face **3004** and/or to extend from both the proximal and distal faces **3002**, **3004**. The stub portions **3010** each define a receptacle **3012** through which the corresponding splines **1624***a-c* can extend to slidingly engage. Also in the illustrated example, each of the instrument drive outputs **2912***a-c* includes a stub shaft **3014** extending or protruding past the proximal face **3002** and configured to mate with a mating shaft receptacle.

[0233] FIG. **31** is an isometric top view of the drive puck **2900**, according to one or more embodiments. The body **2902** is shown in phantom in FIG. **31** so as to better illustrate how the spline couplings **2910***a-c* may be in operative engagement with the instrument drive outputs **2912***a-c*, according to some embodiments. As briefly mentioned above, various gearing and/or mechanisms may be arranged within the body **2902** to operatively intermesh and/or couple each of the spline couplings **2910***a-c* with the associated instrument drive output **2912***a-c*. As illustrated, each of the spline couplings **2910***a-c* includes a drive gear **3020***a-c* arranged within the interior volume **3008** of the body **2902**. Here, each of the drive gears **3020***a-c* is integrally formed with its associated spline coupling **2910***a-c*, but the drive gears **3020***a-c* may be differently connected to their respective spline coupling **2910***a-c*. In addition, each of the instrument drive outputs **2912***a-c* includes a driven gear **3022***a-c*. Here, each of the driven gears **3022***a-c* is integrally formed with its associated instrument drive output **2912***a-c*, but the driven gears **3022***a-c* may be differently connected to its respective instrument drive output **2912***a-c*.

[0234] In some embodiments, the drive puck **2900** may further include first and second idler gears **3024***a* and **3024***b* to operatively couple the first and second spline couplings **2910***a,b* with the first and second instrument drive outputs **2912***a,b*, respectively. In the illustrated embodiment, the first and second idler gears **3024***a,b* are arranged within the body **2902**. As illustrated, the first idler gear **3024***a* interposes and intermeshes with both the first drive gear **3020***a* and the first driven gear **3022***a* of the first instrument drive output **2912***a*. In this manner, rotation of the first spline **1624***a* causes the first spline coupling **2910***a* and the first drive gear **3020***a* to rotate, which causes the first idler gear **3024***a* to rotate and drive the first driven gear **3022***a* and the first instrument drive output **2912***a*. Similarly, the second idler gear **3024***b* interposes and intermeshes with both the drive gear **3020***b* of the second spline coupling **2910***b* and the driven gear **3022***b* of the instrument drive output **2912***b*. In this manner, rotation of the second spline **1624***b* causes the second spline coupling **2910***b* and the second drive gear **3020***b* to rotate, which causes the second idler gear **3024***b* to rotate and drive the second driven gear **3022***b* and the instrument drive output **2912***b*.

[0235] In the illustrated embodiment, the drive puck **2900** also includes a ring gear **3026** arranged to operatively couple the third spline coupling **2910***c* with the third instrument drive output **2912***c*. In other embodiments, however, the third spline coupling **2910***c* and the third instrument drive output **2912***c* may be operatively coupled via a different means, for example, via an idler gear as described above. Where utilized, the ring gear **3026** is arranged to mesh with both the third drive gear **3020***c* of the third spline coupling **2910***c* and the third driven gear **3022***c* of the third instrument drive output **2912***c*. Here, the ring gear **3026** is mounted within the interior volume **3008** of the body **2902** such that the ring gear **3026** is able to rotate relative to the body **2902**. As illustrated, the ring gear **3026** extends continuously along an inner circumferential wall of the body **1902** such that it intermeshes with both the drive gear **3020***c* and the driven gear **3022***c*. In this manner, rotation of the third spline **1624***c* and the third spline coupling **2910***c* rotates the third drive gear **3020***c*, which in turn causes the ring gear **3026** to rotate and drive the driven gear **3022***c* meshed therewith to thereby cause rotation of the third instrument drive output **2912***c*.

[0236] Thus, the drive puck **2900** may be configured to travel along the splines **1624***a-c* and receive the splines **1624***a-c* at respective spline openings located at a distal (e.g., bottom) side of the body **2902**, and transfer the torsional forces (torque) from the splines **1624***a-c* to the corresponding instrument drive outputs **2912***a-c* arranged on an opposite proximal (e.g., top) side of the body **2902** at locations spaced away (or offset) from the associated splines **1624***a-c*.

[0237] FIG. **32** illustrates a portion of the surgical tool **1600** (FIG. **16**), according to one or more embodiments. In the illustrated embodiment, the surgical tool **1600** may be segmented into different portions or segments. For example, the surgical tool **1600** may include a stage segment (FIG. **29**), which incorporates the drive puck **2900** and its functionality, and may further include a handle segment **3200** mountable to the drive puck **2900** and operatively connected to both the shaft **1602** and the end effector **1604** for actuating the same. The handle segment **3200**, the shaft **1602**, and the end effector **1604** may all be integrated together as a unit that may be removed from the stage segment, such that a new unit may be installed on the same stage segment. Accordingly, the handle segment **3200**, the shaft **1602**, and the end effector **1604** may all be integrated together to as a removable segment of the surgical tool **1600**, which may be installed and uninstalled from the stage segment (FIG. **29**) and replaced with a new removable segment.

[0238] In the illustrated embodiment, the handle segment **3200** includes a handle drive housing **3202** having a mounting surface **3204** configured to be mounted on or otherwise mate with the proximal face **3002** (FIG. **30**A) of the drive puck **2900** (FIG. **29**). In particular, the mounting surface **3204** is configured to mate with and engage the instrument drive outputs **2912***a-c* (FIG. **30**A). Thus, the handle drive housing **3202** and the mounting surface **3204** may include geometries corresponding with at least a portion of the drive puck **2900**. The housing **3202** is depicted in FIG. **32** in phantom so as to illustrate the various activating mechanisms contained within the housing **3202**.

[0239] FIG. **33** is an enlarged isometric view of the handle segment **3200** with the housing **3202** shown in phantom to further illustrate exemplary operation, according to one or more embodiments. A plurality of drive inputs **3206***a-c* are provided or otherwise defined in the mounting surface **3204**. In the illustrated example, three drive inputs **3206***a-c* are provided to communicate with and engage the instrument drive outputs **2912***a-c* (FIG. **30**A) of the drive puck **2900** (FIG. **29**), and thereby actuate the end effector **1604**. However, more or less than three drive inputs **3206***a-c* may be provided if the drive puck **2900** includes more or less than three instrument drive outputs **2912***a-c*. As illustrated, each drive input **3206***a-c* defines or otherwise provides a receptacle having a geometry that corresponds to the geometry of the associated instrument drive output **2912***a-c* so as to be able to transfer torsional forces (torque) therebetween.

[0240] The first drive input **3206***a* may be operatively coupled to or form part of the first activating mechanism **1638***a*, which operates to open and close the jaws **1610**, **1612**. Accordingly, rotating the first drive input **3206***a* will correspondingly actuate the first activating mechanism **1638***a* and thereby open or close the jaws **1610**, **1612**, depending on the rotational direction of the first instrument drive output **2912***a*. Similarly, the second drive input **3206***b* may be operatively coupled to the second activating mechanism **1638***b*, which operates to articulate the end effector **1604** at the wrist **1606**. Accordingly, rotating the second drive input **3206***b* will correspondingly actuate the second activating mechanism **1638***b* and cause the wrist **1606** to articulate in at least one degree of freedom, depending on the rotational direction of the second instrument drive output **2912***b*. In addition, the third drive input **3206***c* may be operatively coupled to the third activating mechanism **1638***c*, which operates to fire the cutting element at the end effector **1604**. Accordingly, rotating the third drive input **3206***c* will correspondingly actuate the third activating mechanism **1638***c* and cause the knife to advance or retract, depending on the rotational direction of the third instrument drive output **2912***c*.

[0241] In the illustrated embodiment, the drive inputs **3206***a-c* each include a respective shaft **3210***a-c* that extends proximally into the housing **3202** and is connected to a respective drive gear **3212***a-c*. Also, each drive gear **3212***a-c* is arranged to engage the activating mechanism **1628***a-c* associated with it. Alternatively, as indicated above, the drive gears **3212***a-c* may each form an integral part of the corresponding activating mechanism **1628***a-c*.

[0242] Here, the first drive gear **3212***a* is intermeshed with a driven gear **3214** of the first activating mechanism **1638***a*. The driven gear **3214** may be internally threaded and arranged about an externally threaded portion of the shaft **1602**, such that rotation of the internally threaded driven gear **3214** via rotation of the first drive inputs **3206** threadably engages the externally threaded portion of the shaft **1602** and thereby drives the shaft **1602** axially to open and/or close the jaws **1610**, **1612** (FIG. **32**). The second drive gear **3212***b* may be arranged to engage a pinion or idler gear **3216**, which is operatively intermeshed between the second drive gear **3212***b* and a driven gear **3218** of the third activating mechanism **1638***c* to thereby effectuate firing of the cutting means of end effector **1604**. The third drive gear **3212***c* may be arranged to intermesh with a pair of internally threaded driven gears **3220***a,b* of the second activating mechanism **1638***b*. The pair of internally threaded driven gears **3220***a,b* have opposite thread patterns and are arranged about corresponding threaded carriers **3224***a* and **3224***b*, which translate in opposing axial directions upon rotation of the internally threaded driven gears **3220***a,b*. This causes a pair of linearly translatable internal drive rods **3226***a* (one shown) to antagonistically move distally or proximally. The internal drive rods **3226***a* are operatively coupled to the wrist **1606**, and the antagonistic push and pull of the drive rods operate to control articulation of the wrist **1606**.

[0243] FIG. **34** illustrates an exemplary alignment between the handle segment **3200** and the drive puck **2900** when installing the removable segment of the surgical tool **1600**, according to one or more embodiments. In the illustrated embodiment, the first instrument drive output **2912***a* is alignable and matable with the first drive input **3206***a*, the second instrument drive output **2912***b* is alignable and matable with the second drive input **3206***b*, and the third instrument drive output **2912***c* is alignable and matable with the third drive input **3206***c*. Moreover, the shaft **1602** is extendable through the bore **3006**.

[0244] As discussed above, the drive inputs **3206***a-c* are driven by the instrument drive outputs **2912***a-c*, which are driven indirectly from locations corresponding with the splines **1624***a-c* with internal gearing. In other examples, the drive puck **2900** may include just two instrument drive outputs **2912** driven indirectly from their associated splines with internal gearing. Moreover, in some of these examples, the drive puck **2900** may also be configured to permit direct drive of the handle segment **3200**. For example, the handle segment **3200** may have two or more inputs corresponding with and driven by the indirectly driven elevator outputs of the drive puck **2900**, and the handle segment **3200** may also include one or more inputs each directly activated by an additional spline (unaffiliated with an indirectly driven elevator output), such that two or more inputs of the handle segment **3200** are indirectly driven by the drive puck **2900** and one or more other inputs of the handle segment **3200** are directly driven by splines.

[0245] Accordingly, the drive puck **2900** transfers (or maps) the drive position of the handle segment portion **3200** to locations offset from (or unassociated or unaligned with) the splines **1624***a-c*. While the drive puck **2900** has been described with reference to gearing that mechanically connects the splines **1624***a-c* with the associated instrument drive outputs **2912***a-c*, different means may be utilized to transfer input of the splines **1624***a-c* to the instrument drive outputs **2912***a-c*. For example, the splines **1624***a-c* may be mechanically connected to the instrument drive outputs **2912***a-c* with belts or other mechanisms. Locating the instrument drive outputs **2912***a-c* at drive positions unassociated with the splines **1624***a-c* to which they are mechanically connected provides greater flexibility when designing the geometry of the handle segment portion **3200** and/or the shroud **1640**.

[0246] FIG. **35** illustrates the surgical tool **1600** with the handle segment **3200** installed on the drive puck **2900** of the stage segment. Here, the shaft **1602** is extended through the bore **3006** in the drive puck **2900** and the mounting surface **3204** of the handle drive housing **3202** (FIG. **34**) is positioned on the proximal face **3002** (FIG. **34**) of the drive puck **2900**. In particular, the handle segment **3200** is positioned on the drive puck **2900** such that the drive inputs **3206***a-c* are in alignment with the instrument drive outputs **2912***a-c* of the drive puck **2900**.

Surgical Tool with Removable Instrument Core

[0247] FIG. **36** illustrates another example embodiment of the surgical tool **1600**, according to various embodiments. As mentioned above, the surgical tool **1600** may be assembled by removing the removable cap **1906** and then mounting the handle segment **3200** (FIG. **32**) on the drive puck **2900** (FIG. **29**), which functions as the stage portion **1902** (FIG. **19**) of the surgical tool **1600**, as exemplified in FIGS. **34-35**. In other embodiments, the surgical tool **1600** may be configured such that at least a portion of the instrument portion **1904**

[0248] (FIG. **19**) is an interchangeable sub-assembly (or core) that may be installed on the stage portion **1902**. For example, the handle assembly **1908**, the elongated shaft **1902**, and the end-effector **1604** of the removable instrument portion **1904** may be integrated together as an interchangeable core of the surgical tool **1600** that may be dropped straight onto (or angled onto) the stage portion **1902** without having to remove the removable cap **1906**.

[0249] In FIG. **36**, the surgical tool **1600** includes or otherwise incorporates a handle sub-assembly or handle core **3602** configured to be dropped or angled (or arced) into (i.e., installed in) a stage sub-assembly **3604**. In this illustration, the shroud **1640** (FIG. **17**) has been removed so as to more easily illustrate the stage sub-assembly **3604**. The handle core **3602** (hereinafter, the core **3602**) is an assembly that generally comprises the instrumentation componentry and therapeutic and/or diagnostic features of the surgical tool **1600**. Thus, as illustrated, the core **3602** may include the shaft **1602**, the end effector **1604**, and the operational engagement features configured to affect movement and functionality of the end effector **1604** when engaged by the stage sub-assembly **3604**.

[0250] In the illustrated embodiment, the stage sub-assembly **3604** includes a carriage or kart **3606** configured to translate between the first and second ends **1618***a,b*. Thus, the carriage **3606** may include a carriage nut or kart nut (obscured from view) that is operatively engaged with the lead screw **1622** such that rotation of the lead screw **1622** translates the carriage **3606** along the lead screw **1622**. The splines **1624***a-c* may be arranged within a lower half (e.g., 180°) of the stage sub-assembly **3604** and extend through the carriage **3606**. Thus, the carriage **3606** may include spline channels extending therethrough for receiving the splines **1624***a-c* such that the splines **1624***a-c* may rotate within the spline channels as the carriage **3606** translates along the lead screw **1622**. As more fully described below, spline couplings may be arranged within each of the spline channels, where the spline couplings each receive a corresponding spline **1624***a-c* such that they rotate in unison with their respective spline **1624***a-c*, and the spline couplings are axially constrained within their respective spline channels such that the spline couplings may slide along the splines **1624***a-c* as the carriage **3606** axially translates.

[0251] In addition, the stage sub-assembly **3604** includes a pair of guide rails **3608***a,b* extending between the first and second ends **1618***a,b*. The guide rails **3608***a,b* extend substantially parallel with the splines **1624***a-c* of the stage sub-assembly **3604**, and help facilitate installation of the core **3602** on the carriage **3606**. Moreover, the guide rails **3608***a,b* guide the core **3602** as it is carried by the carriage **3606** when translating proximally or distally between the first and second ends **1618***a,b*. In this example, the guide rails **3608***a,b* are unconstrained by the carriage **3606**. More specifically, the carriage **3606** is generally U-shaped and defines a trough **3610** (or space or void) sized and shaped to receive the core **3602** in a nested relationship and also constrain the core **3602** installed therein such that the carriage **3606** carries the core **3602** as the carriage **3606** translates between the first and second ends **1618***a,b.*

[0252] Also in the illustrated embodiment, the core **3602** includes a pair of arms or wings **3612***a,b* laterally extending from a drive housing or body **3614** of the core **3602**, where the arms **3612***a,b* are configured to be releasably attached to the guide rails **3608***a,b*. In some embodiments, the arms **3612***a,b* rest on the guide rails **3608***a,b*. In some embodiments, the arms **3612***a,b* may be configured to slidingly snap onto the guide rails **3608***a,b* when the body **3614** of the core **3602** is positioned in the trough **3610** in the carriage **3606**, such that the arms **3612***a,b* retain the body **3614** on the rails **3608***a,b* while translating between the first and second ends **1618***a,b*. In some embodiments, the arms **3612***a,b* may include locking toggle levers configured to slidingly attach the arms **3612***a,b* on the guide rails **3608***a,b* when the body **3614** of the core **3602** is positioned in the trough **3610** in the carriage **3606**. In some embodiments, the arms **3612***a,b* include semicircular closure members that rotate and close around the rails **3608***a,b* upon activation of a locking switch. In one non-illustrated embodiment, one or more of the drive splines operate as the guide rail on which the core **3602** is positioned, for example, the arms **3612***a,b* may be attachable on the first and second spline **1624***a,b* that engage a respective drive function, such as the drive gears **3212***a-c* (FIG. **33**).

[0253] FIG. **37** is a schematic diagram depicting example installation of the core **3602** to the stage sub-assembly **3604** of FIG. **36**. As illustrated, the core **3602** may be dropped on or angled into the stage sub-assembly **3604**. As the core **3602** descends, the arms **3612***a,b* may be aligned to receive the guide rails **3608***a,b*, and thus resting the core **3602** on the guide rails **3608***a,b*. In addition, the body **3614** includes operative engagement features for actuating the shaft **1602** (FIG. **36**) and the end effector **1604** (FIG. **36**). In the illustrated examples, the core **3602** is configured to interface with the spline couplings within the carriage **3606** so as to control operation of the end effector **1604**. For example, the carriage **3606** may slide on the splines **1624***a-c* with gears within the carriage driven by and translating on the splines **1624***a-c* that interface with the operative engagement features in the body **3614**, such as the drive gears **3212***a-c* (FIG. **33**).

[0254] FIGS. **38** and **39** illustrate another example of the surgical tool **1600** incorporating a handle sub-assembly or handle core **3802** configured to be dropped into a stage assembly **3902**, according to various embodiments. In FIG. **38**, the core **3802** is illustrated when removed from the stage assembly **3902** (FIG. **39**), whereas FIG. **39** illustrates the stage assembly **3902** without the core **3802** (FIG. **38**) installed therein. The handle core **3802** (hereinafter, the handle **3802**) is an assembly that generally comprises the instrumentation componentry and therapeutic and/or diagnostic features of the surgical tool **1600**. Thus, as illustrated, the core **3802** may include a drive end **3804**, the elongate shaft **1602** which extends from the drive end **3804**, the end effector **1604**, the wrist **1606**, the jaws **1610**,**1612**, and the various operational engagement features configured to affect movement and functionality of the end effector **1604** when engaged by the stage assembly **3902**. The drive end **3804** includes the various drive elements configured to manipulate the various functionality of the end effector **1604**, such as opening and closing of the jaws **1610**,**1612**, articulation of the wrist **1606**, and firing of the end effector **1604**. Here, the drive end **3804** includes the driven gear **3214** of the first activating mechanism **1638***a* (FIG. **32**), the pair of internally

threaded driven gears **3220***a*,*b* of the second activating mechanism **1638***b*, and the driven gear **3218** of the third activating mechanism **1638***c*. As shown in FIG. **39**, the stage assembly **3902** includes a carriage or cart **3904**, configured to translate between the first and second ends **1618***a*,*b* as described herein. Accordingly, the carriage **3904** may be referred to as the elevator. The carriage **3904** includes a central opening or bore **3906**. When the removable cap **1906** is removed, the core **3802** may be dropped into the stage assembly **3902**, with the elongate shaft **1602** being inserted, along the longitudinal axis A.sub.1 (FIG. **16**), through the bore **3906** and through an opening at the first end **1618***a* of the stage assembly **3902** leading into the alignment nozzle **1812** (FIG. **16**), such that the drive end **3804** is positioned within the bore **3906** and thereby operatively coupling the various drive elements of the drive end **3804** with the associated splines **1624***a-c*. In this manner, when the drive end **3804** is engaged within the bore **3906** of the carriage **3904**, the driven gear **3214** of the first activating mechanism **1638***a* will be coupled with the first spline **1624***a*, the pair of internally threaded driven gears **3220***a*,*b* of the second activating mechanism **1638***b* will be coupled with the second spline **1624***b*, and the driven gear **3218** of the third activating mechanism **1638***c* will be coupled with the third spline **1624***c*. Thus, the core **3802** may be installed on the stage assembly **3902** by axially inserting the core **3802** into and through the stage assembly **3902**.
[0255] FIGS. **40** and **41** illustrate another example of the surgical tool **1600** incorporating a handle sub-assembly or handle core **4002** configured to be dropped or angled into a stage sub-assembly **4004**, according to various embodiments. FIG. **40** is an isometric view of a portion of the handle core **4002** (hereinafter, the core **4002**) when removed from the stage sub-assembly **4004**, according to various embodiments, and FIG. **41** is a side view of the portion of the core **4002** and the stage sub-assembly **4004** shown in FIG. **40**.
[0256] In the illustrated embodiment, the stage sub-assembly **4004** includes a carriage or kart **4006** configured to translate between the first and second ends **1618***a*,*b* (FIGS. **36** and **38**). Thus, the carriage **4006** may include a carriage nut **4007** that is operatively engaged with the lead screw **1622** such that rotation of the lead screw **1622** translates the carriage **4006** along the lead screw **1622**. As illustrated, the splines **1624***a-c* extend through the carriage **4006**. Thus, the carriage **4006** may include spline channels extending therethrough for operatively receiving the splines **1624***a-c* such that the splines **1624***a-c* may rotate within the spline channels of the carriage **4006** as the carriage **4006** translates along the lead screw **1622**. In the illustrated embodiment, the splines **1624***a-c* and their corresponding spline channels are arranged within a lower one-hundred and eighty degrees of the stage sub-assembly **4004** and the carriage **4006** to facilitate installation of the core **4002** onto the stage sub-assembly **4004**.
[0257] Drive gears **4008***a-c* may be rotatably mounted to the carriage **4006** and slidingly arranged on each of the splines **1624***a-c* such that the drive gears **4008***a-c* may slide along the respective spline **1624***a-c* while rotating in unison with the associated spline **1624***a-c*. The drive gears **4008***a-c* may be constrained by the carriage **4006** such that they may rotate with their respective spline **1624***a-c* while translating with the carriage **4006**. Here, the drive gears **4008***a-c* are aligned with a corresponding spline channel to slidingly receive their corresponding splines **1624***a-c*. The drive gears **4008***a-c* rotate in unison with their respective spline **1624***a-c* while being axially constrained relative to the carriage **4006** such that they may slide over the splines **1624***a-c* as the carriage **4006** translates. Thus, the first drive gear **4008***a* is configured to slide axially along while rotating in unison with the first spline **1624***a*, the second drive gear **4008***b* is configured to slide axially along while rotating in unison with the second spline **1624***b*, and the third drive gear **4008***c* is configured to slide axially along while rotating in unison with the third spline **1624***c*.
[0258] The carriage **4006** may be configured to provide a trough **4010** (or space or void) within which the core **4002** may be mounted. In the illustrated example, the carriage **4006** defines a trough **4010** sized and shaped to receive the core **4002**. Here, the carriage **4006** constrains the core **4002** once mounted in the trough **4010**, such that the carriage **4006** carries the core **4002** as it translates. While the carriage **4006** is illustrated having a generally U-shaped geometry, it may exhibit

different geometries, without departing from the present disclosure. For example, the carriage **4006** may include a circular shaped opening for receiving a drive end with drive elements of a drop-in core, such as the core **3802** described with reference to FIGS. **38-39**.

[0259] The drive gears **4008***a-c* of the carriage **4006** are each configured to drive,

[0260] engage, and interact with a corresponding activating mechanism **4012***a-c* of the core **4002**. The activating mechanisms **4012***a-c* may be arranged at various locations along the axial length of a drive housing or body **4014** of the core **4002**. In the illustrated example, the first activating mechanism **4012***a* is located at or near a proximal end of the body **4014**, the third activating mechanism **4012***c* is located at or near a distal end of the body **4014**, and the second activating mechanism **4012***b* is located between the first and third activating mechanisms **4012***a,c*. As will be appreciated, the activating mechanism **4012***a-c* may be located differently, depending on the application.

[0261] In some embodiments, one or more of the drive gears **4008***a-c* (and/or their teeth or cogs) may extend into the trough **4010** to engage (intermesh) the corresponding activating mechanism **4012***a-c* of the core **4002**. In other embodiments, however, one or more of the drive gears **4008***a-c* may be recessed within the body of the carriage **4006** and the corresponding activating mechanisms **4012***a-c* may extend to engage the drive gears **4008***a-c*. In the illustrated example, the first drive gear **4008***a*, which is slidingly arranged on the first spline **1624***a*, is configured to mesh with the first activating mechanism **4012***a* such that rotation of the first spline **1624***a* correspondingly actuates the first activating mechanism **4012***a* and thereby carries out a first function of the surgical tool **1600** (e.g., to fire a cutting element). The second drive gear **4008***b*, which is slidingly arranged on the second spline **1624***b*, is configured to mesh with the second activating mechanism **4012***b* such that rotation of the second spline **1624***b* correspondingly actuates the second activating mechanism **4012***b* and thereby carries out a second function of the surgical tool **1600** (e.g., to articulate the wrist **1606** of FIG. **35**). Lastly, the third drive gear **4008***c*, which is slidingly arranged on the third spline **1624***c*, is configured to mesh with the third activating mechanism **4012***c* such that rotation of the third spline **1624***c* correspondingly actuates the third activating mechanism **4012***c* and thereby carries out a third function of the surgical tool **1600** (e.g., to open or close the jaws **1610**, **1612** of FIG. **35**).

[0262] Various means may be utilized to retain the body **4014** of the core **4002** within the carriage **4006** of the stage sub-assembly **4004**. In some examples, the carriage **4006** includes a pair of sidewalls **4016***a,b* on either side of the trough **4010**, and the sidewalls **4016***a,b* may be biased or otherwise angled inward so as to retain the body **4014** when the body **4014** is inserted therein. In such embodiments, the core **4002** may be received within the trough **4010** via a snap-fit or interference fit engagement. However, other means may be utilized to releasably secure the body **4014** to the carriage **4006**, for example, various types of mechanical fasteners, snaps, magnets, Velcro, etc. In other embodiments, a closure band or shroud portion may rotate about the shroud **1640** (FIG. **35**) to constrain and enclose the body **4014** on the carriage **4006**.

[0263] FIGS. **42**A-**42**D depict various alternative embodiments of the shroud **1640** that may form part of the surgical tool **1600** of FIG. **16**. More specifically, FIGS. **42**A-**42**D illustrate example shroud assemblies configured to open and close such that a handle core or handle sub-assembly (e.g., the core **3602**, **3802**, **4002** of FIGS. **36-37**, FIGS. **38-39**, and/or FIGS. **40-41**, respectively) may be easily installed on a stage sub-assembly (e.g., the stage sub-assembly **3604**, **3804**, **4004** of FIGS. **36-37**, FIGS. **38-39**, and/or FIGS. **40-41**, respectively).

[0264] FIG. **42**A illustrates a shroud assembly **4200** wherein the shroud **1640** thereof defines an opening **4202** configured to receive a handle sub-assembly, according to one or more embodiments. In the illustrated example, the opening **4202** in the shroud **1640** corresponds to the shape of a handle sub-assembly intended to be inserted (introduced) into the shroud **4204**. Thus, the opening **4202** includes an enlarged portion **4204** corresponding in size and shape to the body of the handle sub-assembly (e.g., the body **3614**, **3814**, **4014** of FIGS. **36-37**, FIGS. **38-39**, and FIGS. **40-41**, respectively) and a narrow portion **4206** that extends distally from the enlarged portion and is sized and shaped to receive the shaft **1602** (FIGS. **36**, **38**, and **40**) and the end effector **1604** (FIGS. **36**, **38**, and **40**) of the handle sub-assembly. Accordingly, the handle sub-assembly may be dropped or angled into the stage sub-assembly through the opening **4202**.

[0265] In some embodiments, a secondary shroud may be provided to selectively cover the opening **4202**, such that the handle sub-assembly may be fully enclosed and/or sealed from the ambient environment when installed on the stage sub-assembly. For example, FIGS. **42**B-**42**D illustrate different embodiments of the shroud assembly **4200** having a secondary shroud configured to selectively enclose and seal the opening **4202** in the shroud **1640**.

[0266] FIG. **42**B illustrates another embodiment of the shroud assembly **4200**, according to one or more embodiments. In particular, FIG. **42**B illustrates an example of the shroud assembly **4200** having a secondary shroud **4210** configured to slide over the opening **4202** in the shroud **1640**. In the illustrated embodiment, the shroud **1640** is generally cylindrical in shape and the secondary shroud **4210** is an arcuate member having a curvature substantially similar to the curvature of the shroud **1640** and sized to occlude the opening **4202** in the shroud **1640**. In some embodiments, the secondary shroud **4210** is provided with a curvature that is generally concentric with the shroud **1640** when the secondary shroud **4212** is slidingly attached (or coupled) to the shroud **1640**.

[0267] The secondary shroud **4210** may be able to move (slide) relative to the inner or outer circumference of the shroud **1640**. In the illustrated embodiment, the secondary shroud **4212** is slidingly disposed over an exterior surface (i.e., the outer circumference) of the shroud **1640** such that the secondary shroud **4212** may circumferentially revolve (rotate) over the shroud in a closure direction C; however, in other embodiments, the secondary shroud **4212** may be slidingly disposed within an interior (i.e., the inner circumference) of the shroud **1640** such that the secondary shroud **4212** circumferentially revolves (rotates) within the shroud **1640**. In this manner, the secondary shroud **4210** may overlap a portion of the shroud **1640** by sliding on the perimeter or circumference of the shroud **1640** and thereby expose or occlude the opening **4202**.

[0268] In some embodiments, the secondary shroud **4210** may slide or revolve on the shroud **1640** between an open position, where the opening **4202** is at least partially exposed, and a closed position, where the secondary shroud **4210** occludes the opening **4202**. Here, the secondary shroud **4210** is shown in an open position such that the opening **4202** is exposed, but the secondary shroud **4210** may be angularly rotated and slid circumferentially relative to the shroud **1640** in the closure direction C to cover the opening **4202**. Thus, the secondary shroud **4210** may be circumferentially revolvable (rotatable) relative to the shroud **1640** between open and closed positions.

[0269] FIG. **42**C illustrates another embodiment of the shroud assembly **4200**, according to one or more embodiments. In particular, FIG. **42**C illustrates an example of the shroud assembly **4200** having a secondary shroud **4212** pivotably attached to the shroud **1640**. More specifically, the secondary shroud **4212** may be pivotably attached to the shroud **1640** at a hinge **4213** that allows the secondary shroud **4212** to pivot or rotate relative to the shroud **1640** between open and closed positions. The hinge **4213** pivotably attaches the secondary shroud **4212** to the shroud **1640** at a distal location of the shroud **1640** such that the secondary shroud **4212** may be pivoted upward to an open position, where at least a portion of the opening **4202** is sufficiently exposed for dropping or angling the handle sub-assembly into the opening **4202**. Thereafter, the secondary shroud **4212** may be pivoted downward into a closed position to occlude and possibly seal the opening **4202**. In the illustrated embodiment, the secondary shroud **4212** is shown in an open position, where the opening **4202** is exposed for the handle sub-assembly to be angled or dropped there-into, but the secondary shroud **4212** may be pivoted downward in the closure direction C to thereby close the shroud assembly **4200** and cover the opening **4202**.

[0270] FIG. **42**D illustrates another example embodiment of the shroud assembly **4200** having a secondary shroud **4214** configured to slide axially over the shroud **1640** to expose or occlude the opening **4202**, according to one or more embodiments. In the illustrated embodiment, the secondary shroud **4214** is a tubular-shaped member arranged to slide axially over the shroud **1640**. In particular, the secondary shroud **4214** may slide proximally into an open position where the opening **4202** is at least partially exposed, or the secondary shroud **4214** may slide distally into a closed position where the secondary shroud **4214** at least partially covers the opening **4202**. In the illustrated embodiment, the secondary shroud **4214** is axially slidable over an external surface of the shroud **1640**, but in other embodiments, the secondary shroud **4214** may be axially slidable (telescope) within an interior of the shroud **1640**. Here, the secondary shroud **4214** is illustrated in an open position, where the opening **4202** is exposed for the handle sub-assembly to be angled or dropped there-into, but the secondary shroud **4214** may be axially slid in the closure direction C to thereby close the shroud assembly **4200** and cover the opening **4202**. Thus, the secondary shroud **4214** may be configured to be axially slidable (or telescoping) relative to the shroud **1640** between open and closed positions.

4. Implementing Systems and Terminology

[0271] Implementations disclosed herein provide systems, methods and apparatus for instruments for use with robotic systems. It should be noted that the terms “couple,” “coupling,” “coupled” or other variations of the word couple as used herein may indicate either an indirect connection or a direct connection. For example, if a first component is “coupled” to a second component, the first component may be either indirectly connected to the second component via another component or directly connected to the second component.

[0272] The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is required for proper operation of the method that is being described, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

[0273] As used herein, the term “plurality” denotes two or more. For example, a plurality of components indicates two or more components. The term “determining” encompasses a wide variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” can include resolving, selecting, choosing, establishing and the like.

[0274] The phrase “based on” does not mean “based only on,” unless expressly specified otherwise. In other words, the phrase “based on” describes both “based only on” and “based at least on.”

[0275] As used herein, the terms “generally” and “substantially” are intended to encompass structural or numeral modification which do not significantly affect the purpose of the element or number modified by such term.

[0276] To aid the Patent Office and any readers of this application and any resulting patent in interpreting the claims appended herein, applicants do not intend any of the appended claims or claim elements to invoke 35 U.S.C. 112(f) unless the words “means for” or “step for” are explicitly used in the particular claim.

[0277] The foregoing previous description of the disclosed implementations is provided to enable any person skilled in the art to make or use the present invention. Various modifications to these implementations will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the scope of the invention. For example, it will be appreciated that one of ordinary skill in the art will be able to employ a number corresponding alternative and equivalent structural details, such as equivalent ways of fastening, mounting, coupling, or engaging tool components, equivalent mechanisms for producing particular actuation motions, and equivalent mechanisms for delivering electrical energy.

Thus, the present invention is not intended to be limited to the implementations shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A method of assembling a surgical tool, comprising: inserting an end effector and a shaft of an instrument portion of the surgical tool into an opening of an elevator layer of a stage portion of the surgical tool; extending a plurality of legs of an alignment table through a corresponding plurality of spline passages of a handle assembly of the instrument portion; rotating the instrument portion with the alignment table relative to the stage portion, and thereby aligning the plurality of spline passages with a corresponding plurality of splines penetrating the elevator layer; advancing the handle assembly toward the elevator layer and thereby receiving the plurality of splines within the plurality of spline passages; coupling the handle assembly to the elevator layer to form a carriage; and removing the alignment table from the handle assembly.
2. The method of claim 1, wherein removing the alignment table from the handle assembly comprises: retracting the carriage into a shroud forming part of the stage portion; and ejecting the alignment table from the handle assembly as an end of each spline contacts a corresponding end of each leg.
3. The method of claim 2, further comprising: securing an end cap to a proximal end of the stage portion and the shroud after removing the alignment table; and inserting an end of one or more of the plurality of splines within a corresponding one or more spline couplings of the end cap.
4. The method of claim 1, wherein the plurality of legs includes an additional leg alignable with a screw passage of the handle assembly, and wherein rotating the instrument portion with the alignment table further comprises aligning the additional leg with the screw passage and with a lead screw penetrating the elevator layer via the screw passage.
5. The method of claim 1, wherein coupling the handle assembly to the elevator layer comprises engaging a bulbous portion of a plurality of snaps of the handle assembly with a corresponding plurality of openings formed in the elevator layer.
6. The method of claim 1, wherein extending the plurality of legs through the corresponding plurality of spline passages aligns a plurality of layers of the handle assembly.
7. The method of claim 6, wherein the plurality of layers include one or more activating mechanisms, the method further comprising rotating at least one of the plurality of splines and thereby actuating the one or more activating mechanisms to operate the surgical tool.
8. A method of assembling a surgical tool, comprising: moving an elevator layer of a stage portion of the surgical tool to a first position in which the elevator layer partially protrudes beyond an end of a shroud of the stage portion; extending a plurality of legs of an alignment table through a corresponding plurality of spline passages of a handle assembly of an instrument portion of the surgical tool, and thereby aligning a plurality of layers of the handle assembly, the plurality of spline passages being defined in the plurality of layers; coupling the handle assembly to the elevator layer in the first position and thereby forming a carriage; moving the carriage from the first position to a second position and thereby receiving the carriage completely within the shroud; removing the alignment table from the handle assembly as the carriage moves to the second position; and attaching an end cap to the end of the shroud after moving the elevator layer to the second position.
9. The method of claim 8, wherein removing the alignment table from the handle assembly comprises: contacting each leg with an end of a corresponding one of the plurality of splines as the carriage moves from the first position toward the second position; and pushing the plurality of legs out of the corresponding plurality of passages as the carriage continues toward the second position.
10. The method of claim 8, wherein two or more of the plurality of legs exhibit a different length.
11. The method of claim 8, wherein coupling the handle assembly to the elevator layer comprises

engaging a bulbous portion of a plurality of snaps of the handle assembly with a corresponding plurality of openings formed in the elevator layer.

12. The method of claim 11, further comprising radially restraining the bulbous portion of each snap within the corresponding plurality of openings with the shroud when the elevator layer is in the second position.

13. The method of claim 8, wherein coupling the handle assembly to the elevator layer in the first position is preceded by: rotating the instrument portion with the alignment table relative to the stage portion; and aligning the plurality of spline passages with a corresponding plurality of splines penetrating the elevator layer.

14. The method of claim 13, wherein the carriage incudes one or more activating mechanisms, the method further comprising rotating at least one of the plurality of splines and thereby actuating the one or more activating mechanisms to operate the surgical tool.

15. A method of assembling a surgical tool, comprising: removing an end cap from a stage portion of the surgical tool, the stage portion including: opposing first and second ends, the end cap being removed from the second end; a lead screw and a plurality of splines extending between the first and second ends; and a first layer of a carriage movably mounted to the lead screw and the plurality of splines; advancing the first layer to the second end; coupling an instrument portion to the stage portion, the instrument portion including: a plurality of additional layers of the carriage configured to be removably coupled to the first layer, the plurality of additional layers being aligned together by an alignment table having a plurality of legs disposed in corresponding passages formed in the plurality of additional layers, each passage further corresponding to either the lead screw or a spline of the plurality of splines; and an elongate shaft extending distally from the one or more additional layers and having an end effector arranged at a distal end of the elongate shaft, the elongate shaft being disposed within an opening in the first layer when the instrument portion is coupled to the stage portion; and removing the alignment table from the plurality of additional layers.

16. The method of claim 15, wherein removing the alignment table from the plurality of additional layers comprises ejecting the alignment table from the plurality of additional layers in response to moving the first layer and the plurality of additional layers coupled thereto toward the first end to contact each leg of the alignment table with a corresponding spline of the plurality of splines.

17. The method of claim 15, further comprising reattaching the end cap to the second end of the stage portion after removing the alignment table.

18. The method of claim 15, wherein the surgical tool includes a shroud within which the carriage translates, and wherein advancing the first layer to the second end comprises: advancing the first layer until one or more openings defined about an outer periphery of the first layer extend outside of the shroud; and receiving one or more snaps extending from the plurality of additional layers at the one or more openings when the one or more openings are positioned outside of the shroud, thereby coupling the instrument portion to the stage portion.

19. The method of claim 18, further comprising: advancing the carriage toward the first end with the one or more snaps received within the one or more openings; and constraining the one or more snaps within the one or more openings with the shroud, and thereby securing the instrument portion to the stage portion.

20. The method of claim 15, wherein the plurality of legs exhibit differing lengths.
